

ENGINEERING TEAMS / RESEARCH TO PRODUCTION

AI Development English

Precise language for models, evidence,
and delivery.

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Eight lessons. Eleven extended conversations.

INSIDE THE BOOK

192 vocabulary entries / 128 reusable phrases
220 extended-dialogue turns / 11 transfer exchanges
Word banks and explained answer keys

Upper-intermediate to advanced (B2-C1). For AI engineers, researchers, technical leads, and product colleagues.

2026 edition

Eight scenarios. Eight skills.

Explain the system, question the evidence, and agree on a defensible next step.

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One proposal, two different failure modes | Separate proposed remedies by mechanism and agree on a controlled experiment.

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Name the claim. Name the evidence.

AI development discussions move between research, software, product, and risk. Shared words do not always carry shared meanings. Effective English makes the component, comparison, and uncertainty explicit.

Separate the system from the model

An application can fail in retrieval, prompting, serving, permissions, or presentation even when its model is unchanged.

"The retrieved passage is correct; the generated answer omits the exception."

Keep the denominator visible

A percentage needs a population and criterion. An average can conceal weak performance on an important slice.

"It passed 88 of these 100 cases; that is not an estimate for all customer traffic."

Disagree with a testable claim

Ablations, baselines, and matched comparisons make technical disagreement useful. A request for evidence is not necessarily a personal challenge.

"What changes if we hold the retrieval set constant and swap only the reranker?"

Do not turn a hypothesis into a finding

Use may, appears, and is consistent with when evidence is limited. State what would confirm or weaken the explanation.

"The timing suggests a queueing issue, but we have not isolated the cause."

PROFESSIONAL BOUNDARIES

Fictional communication practice, not deployment, security, or compliance instructions. Use authorized review, current documentation, and approved procedures for real systems. Never place real secrets or personal data in practice exercises.

The answer is wrong, but which layer failed?

A Read the situation

Distinguish components and ask a diagnostic question without assigning blame.

A document assistant displays the wrong refund deadline. Retrieval returned the current policy, which states 30 days, but the generated answer says 60. The interface displays the generated text unchanged. Priya owns the model integration; Alex manages the product. A demonstration is scheduled for Friday. No cause has been confirmed, and Alex cannot approve a production release alone.

PEOPLE / Priya: AI integration engineer | Alex: Product manager

LOCATE THE FAILURE BEFORE NAMING AN OWNER

"The AI is broken" does not identify a component or establish responsibility. Separate the observed output from a theory about its cause. A colleague asking for the exact trace is asking for evidence, not necessarily dismissing the report.

Briefing checks

Circle one answer in each item. Use the briefing.

1. Which observation is confirmed?

- | | |
|--|---|
| A Production release is approved. | B The generated answer says 60 days. |
| C Retrieval found an obsolete policy. | D The interface altered the number. |

2. Who owns the model integration?

- | | |
|----------------------------|-----------------------|
| A The policy author | B Priya |
| C Alex alone | D The customer |

3. What remains unknown?

- | | |
|--|--|
| A Whether a demonstration is scheduled | B Whether the interface displays the text |
| C Why the model produced the wrong deadline | D The correct policy deadline |

Answers and explanations: [page 85](#).

The vocabulary that does the work

B Find the words / 1 of 2

Read the definitions and say the common word combinations aloud.

application stack

The components that together deliver an application.

IN USE map the application stack

retrieval

Finding source material relevant to a request.

IN USE inspect retrieval results

language model

A model that processes and generates language sequences.

IN USE query a language model

user interface

The part of a system a user sees and operates.

IN USE inspect the user interface

orchestration

Coordination of model, retrieval, and tool steps.

IN USE trace the orchestration flow

inference

Running a trained model to produce an output.

IN USE run inference

endpoint

An address through which a service accepts requests.

IN USE call an endpoint

payload

The data carried in a request or response.

IN USE inspect the request payload

trace

A linked record of events in a request's path.

IN USE examine a request trace

dependency

A component or condition another part relies on.

IN USE identify a service dependency

upstream

Earlier in the relevant processing path.

IN USE inspect the upstream response

downstream

Later in the relevant processing path.

IN USE assess downstream impact

Precision check

A trace records events; a diagnosis explains them. A wrong answer identifies a symptom, not automatically a model defect. Keep retrieval output, model output, and displayed text separate.

The vocabulary that does the work

B Find the words / 2 of 2

Read the definitions and say the common word combinations aloud.

symptom

An observed sign of a problem, not its cause.

IN USE describe the symptom

root cause

The underlying cause established by investigation.

IN USE verify the root cause

reproduction

Recreating a reported result under stated conditions.

IN USE obtain a reliable reproduction

request ID

An identifier used to locate a particular request.

IN USE attach the request ID

schema

A defined structure for data and its fields.

IN USE validate a response schema

tool call

A model-requested invocation of an available tool.

IN USE inspect a tool call

API

Application programming interface; a defined software interaction boundary.

IN USE document an API contract

contract

Agreed expectations between software components.

IN USE preserve the service contract

integration

Connecting components into a working system.

IN USE test the integration

observability

The ability to infer system behavior from available signals.

IN USE improve observability

triage

Initial assessment that determines priority and ownership.

IN USE triage the defect

handoff

Transfer of work and its relevant context.

IN USE prepare a clear handoff

Precision check

An API contract defines expected interaction, not necessarily a legal contract. "Upstream" and "downstream" describe position relative to a chosen component; name that component.

Phrases you can actually use

C Notice the language / 1 of 2

LOCATE THE LAYER

"Which component first shows the wrong value?"

LIMIT THE CONCLUSION

"That locates the mismatch, but it does not establish the cause."

SEPARATE RESPONSIBILITIES

"Product approval and technical diagnosis are different decisions."

CLARIFY A TERM

"By 'the AI,' do you mean the model or the whole application?"

STATE THE OBSERVATION

"The retrieved passage says 30 days; the generated answer says 60."

REQUEST A TRACE

"Can you attach the request ID and the redacted trace?"

ASK FOR REPRODUCTION

"Can we reproduce it with the same payload and configuration?"

AGREE ON A HANDOFF

"I will send the evidence and name the unresolved question."

Which, not why

"Which component first shows the mismatch?" asks for location before assuming a cause.

Present evidence

"The trace shows" introduces an observation; "this suggests" introduces an interpretation.

Embedded questions

Say "Can you confirm whether the interface changes it?", not "whether does the interface change it?"

Phrases you can actually use

C Notice the language / 2 of 2

CHECK THE INTERFACE

"Does the interface display the response without changing it?"

NARROW THE TEST

"Change one component while keeping the others fixed."

CONFIRM A DEPENDENCY

"Does this test depend on the policy index being current?"

SET THE UPDATE

"I can commit to an update, not to a fix by that time."

AVOID BLAME

"Let us inspect the boundary before assigning ownership."

NAME THE UNCERTAINTY

"We have an observation, not a verified root cause."

PRESERVE THE RECORD

"Please retain the configuration used for that request."

CLOSE PRECISELY

"The demonstration remains scheduled; release approval is separate."

Contrast clauses

Use "retrieval returned the right passage, but generation changed the deadline" to isolate the contrast.

Limited commitments

"I will investigate by Friday" is unclear; "I will report our findings by Friday" names a deliverable.

Ownership language

"I own the investigation" accepts responsibility for the work, not blame for the incident.

Choose the precise message

D Check your understanding

Circle one best answer for each item. Use the stated purpose and case facts, not the option that merely sounds most polite.

1. Which question locates the first mismatch?

- A** Should the interface team own the cause because it displays the answer?
- B** Can a single corrected reply establish that the issue is fixed?
- C** At which boundary does 30 become 60?
- D** Can we assign the fault to the model because retrieval found the right passage?

2. Complete: "Could you confirm whether ___?"

- A** the payload does includes the policy
- B** the payload includes the policy
- C** does the payload include the policy
- D** includes the payload the policy

3. Which statement preserves uncertainty?

- A** The trace suggests a generation issue; the cause is unconfirmed.
- B** The trace proves the engineer caused the defect.
- C** A correct passage guarantees a correct answer.
- D** The demonstration authorizes production use.

4. Which expression means transferring work with context?

- A** prepare a clear handoff
- B** increase the inference
- C** approve a symptom
- D** retrieve the ownership

[Answers and explanations: page 85.](#)

The answer is wrong, but which layer failed?

E Conversations / turns 1-10

Complete the numbered gaps. Use each bank entry once, without changing its form. Cover the answers at the end until you finish.

WORD + PHRASE BANK

configuration / request ID / triage / mismatch / payload / upstream / handoff / reproduction / root cause / trace

- 01 **Alex:** The refund answer is wrong again. The policy says 30 days, but our assistant tells the customer 60. Which team should investigate first?
- 02 **Priya:** Do you have the ¹ _____? I want to see the retrieved passage and the answer side by side.
- 03 **Alex:** Yes, the record is D-184. I checked the passage in that record: it says 30 days. The response underneath it says 60.
- 04 **Priya:** That gives us a specific ² _____. Retrieval found the current passage. I still need to inspect exactly what reached the model.
- 05 **Alex:** So the model got it wrong? I need to tell the team who is picking this up.
- 06 **Priya:** Tell them the answer is wrong, but leave the cause open. The ³ _____ could contain another instruction or an older example.
- 07 **Alex:** Wait, could the interface have changed the number? We had a display issue last time.
- 08 **Priya:** The recorded response already says 60, and the interface displays it unchanged. That puts the first observed difference ⁴ _____ of presentation.
- 09 **Alex:** What would you need to distinguish a bad instruction from an unstable model answer? We should keep the Friday demonstration team informed.
- 10 **Priya:** The configuration and a redacted ⁵ _____. Then I can repeat the request and compare the inputs without changing several variables.

The conversation, continued

E Conversations / turns 11-20

Continue with the word bank on page 10.

- 11 **Alex:** Is repeating one request enough to say we have fixed the problem? The same customer asked two slightly different versions of the question.
- 12 **Priya:** No. First we need a reliable ⁶ _____. Then we test the change on both variants and related policy cases.
- 13 **Alex:** Please make the ownership clear in the ticket. I can coordinate the demonstration, but I cannot approve production use on my own.
- 14 **Priya:** I will own technical ⁷ _____. I will also name the policy and interface owners if the evidence requires their input.
- 15 **Alex:** Can I tell the sales team we will have a fix by Thursday afternoon? They need to decide whether to keep the live demonstration.
- 16 **Priya:** I can promise findings by then, not a fix. Until we verify a ⁸ _____, that deadline would be an unsupported commitment.
- 17 **Alex:** All right. I will describe the customer-visible problem and keep the demonstration decision separate. What should your update include for that decision?
- 18 **Priya:** The reproduction result, affected cases, remaining uncertainty, and a proposed next step. I will preserve the ⁹ _____ so the comparison is meaningful.
- 19 **Alex:** That works. Please send it by three on Thursday, even if the investigation is incomplete. I will confirm who can decide about Friday.
- 20 **Priya:** Agreed. The ¹⁰ _____ will say what we know, what we do not know, and which decision still needs an owner.

ANSWERS / EXPLANATIONS: PAGE 86

1. request ID | 2. mismatch | 3. payload | 4. upstream | 5. trace | 6. reproduction | 7. triage | 8. root cause | 9. configuration | 10. handoff

A missing citation in a different assistant

F Say it

A support assistant generates a correct answer with a citation ID. The interface omits the citation link. The display team needs the exact recorded request, not an assumption that retrieval failed.

Complete the four gaps. Use each bank entry once.

WORD + PHRASE BANK

symptom / user interface / handoff / request ID

Engineer: "The model response contains the citation; the ¹ _____ does not display it."

Lead: "Attach the ² _____ so the display team can locate that exchange."

Engineer: "That is the observed ³ _____; we have not proved why it occurred."

Lead: "Then send a clear ⁴ _____ to the display owner."

ANSWERS / EXPLANATIONS: PAGE 86

1. user interface | 2. request ID | 3. symptom | 4. handoff

Rehearse with a fixed script

1. Check the dialogue, then read the 30-day source and 60-day generated answer as separate observations. Locate the mismatch before assigning its cause.
2. Switch roles. Ask for D-184, the redacted trace and the request configuration; retain the Thursday findings update without promising a fix.
3. Complete the citation-link exchange. State that the generated response contains the citation while the interface omits it, then hand over the exact request.

WORKING ALONE

Read both roles, pausing for two seconds when the speaker changes. Repeat with the bank covered, then check any missed expressions.

A larger answer will not make the input fit

A Read the situation

Correct a technical misconception tactfully and negotiate a bounded alternative.

A document-summary request reserves 4,000 output tokens. Its instructions and documents contain 29,000 input tokens. The configured total context limit is 32,000 tokens, and the request is rejected before generation. Lina proposes increasing the output allowance. Omar owns the request configuration. The team must retain the exception clauses; no larger-context model has been evaluated for this task.

PEOPLE / Lina: Application engineer | Omar: Model platform engineer

CORRECT THE MODEL OF THE PROBLEM

Restate the proposal before rejecting it. Explain which limit applies and show the arithmetic. "That would increase the reservation" is more useful than "You do not understand tokens." A correction can be direct without judging the colleague.

Briefing checks

Circle one answer in each item. Use the briefing.

1. What is the total requested reservation?

A 29,000 tokens

B 32,000 tokens

C 25,000 tokens

D 33,000 tokens

2. Why was the request rejected?

A The exception clauses were false.

B A larger model had already failed evaluation.

C It exceeds the configured total context limit.

D The generated answer was inaccurate.

3. What must the team preserve?

A A 4,000-token answer in every case

B The untested larger model

C The exception clauses

D Every duplicate header

Answers and explanations: page 87.

The vocabulary that does the work

B Find the words / 1 of 2

Read the definitions and say the common word combinations aloud.

LLM

Large language model; a model that processes and generates language.

IN USE evaluate an LLM

transformer

A neural architecture built around attention mechanisms.

IN USE use a transformer architecture

token

A unit processed by a model, not necessarily a whole word.

IN USE count input tokens

tokenizer

The component that converts text into tokens.

IN USE use the model's tokenizer

context window

The amount of context a model can handle in a request.

IN USE stay within the context window

input tokens

Tokens supplied to the model before generation.

IN USE measure input tokens

output tokens

Tokens generated by the model.

IN USE reserve output tokens

token budget

An allocated number of tokens for a task or component.

IN USE allocate a token budget

system instruction

Higher-priority guidance supplied by an application's configuration.

IN USE inspect the system instruction

prompt

Instructions and other context supplied for a model response.

IN USE revise the prompt

attention

A mechanism that weights relationships between representations.

IN USE compute attention weights

parameter

A learned numerical value within a model.

IN USE update model parameters

Precision check

A token is not a word. Input size, reserved output, and total context are different quantities. The numerical limits in this case are fictional configuration facts, not specifications for a named model.

The vocabulary that does the work

B Find the words / 2 of 2

Read the definitions and say the common word combinations aloud.

weight

A learned numerical coefficient in a model.

IN USE load model weights

checkpoint

Saved model state from a particular stage.

IN USE select a checkpoint

truncation

Removal of content to fit a size limit.

IN USE detect silent truncation

compression

Reduction of information size, potentially losing detail.

IN USE assess compression losses

reserved capacity

Capacity set aside rather than already consumed.

IN USE reduce reserved capacity

delimiter

A marker separating sections of supplied text.

IN USE preserve clear delimiters

instruction hierarchy

Priority relationships among different instruction sources.

IN USE respect the instruction hierarchy

long-context

Able to handle relatively large input context.

IN USE evaluate long-context performance

position

A token's place within the input sequence.

IN USE vary evidence position

exception clause

Wording that limits or changes a general rule.

IN USE retain the exception clause

deduplication

Removal of repeated copies of content.

IN USE apply exact deduplication

headroom

Unused capacity left below a limit.

IN USE leave sufficient headroom

Precision check

Deduplication removes repeated material; summarization changes its representation. Neither guarantees that every important exception remains visible. A larger context window does not by itself establish task accuracy.

Phrases you can actually use

C Notice the language / 1 of 2

RESTATE A PROPOSAL

"You are proposing a larger output allowance, correct?"

SHOW THE ARITHMETIC

"Twenty-nine thousand plus four thousand is thirty-three thousand."

DISTINGUISH QUANTITIES

"The allowance is a maximum, not the length already generated."

OFFER AN ALTERNATIVE

"Could we remove exact duplicates before shortening substantive text?"

LOCATE THE LIMIT

"The request fails before generation because the combined reservation is too large."

CORRECT TACTFULLY

"I see the aim, but that change would increase the reservation."

PRESERVE A REQUIREMENT

"We must retain the exception clauses in the source context."

STATE A CONDITION

"A smaller output allowance is acceptable only if the summary still meets the task."

Arithmetic in speech

Say "thirty-three thousand in total" after stating the two parts; listeners can check the sum.

Would versus will

"That would increase the reservation" describes the proposed change without claiming it has happened.

Only if

"Acceptable only if coverage is preserved" makes coverage a necessary condition.

Phrases you can actually use

C Notice the language / 2 of 2

CHECK COUNTING

"Are those counts from the tokenizer used by this model?"

CHECK A TRANSFORMATION

"Which information would compression remove?"

BOUND A COMPARISON

"Let us compare the same documents under both configurations."

SEPARATE EVALUATION

"The larger-context model needs its own quality check."

AVOID A GUARANTEE

"Fitting within the limit does not prove the answer is complete."

REJECT SILENT LOSS

"Please report truncation instead of silently dropping the end."

KEEP A FALLBACK

"If the exceptions cannot fit, route the request for a different workflow."

READ BACK THE DECISION

"We will remove duplicates, recount, and test coverage before changing models."

Not necessarily

"A larger window is not necessarily more accurate" rejects a guarantee without claiming it is always worse.

Scope of all

"All exceptions in these documents" is bounded; "all exceptions" can imply knowledge beyond the supplied material.

Clarifying by

"By output allowance, do you mean the maximum reserved tokens?" checks a term before challenging it.

Choose the precise message

D Check your understanding

Circle one best answer for each item. Use the stated purpose and case facts, not the option that merely sounds most polite.

1. Which response explains the rejection?

- A** The combined reservation is 1,000 tokens over the limit.
- B** The output has already used 33,000 tokens.
- C** The model generated an incorrect exception.
- D** The tokenizer guarantees the summary is complete.

2. Which proposal preserves the stated requirement most directly?

- A** Remove exact duplicate headers and recount.
- B** Delete exception clauses first.
- C** Increase the output reservation without recounting.
- D** Treat words and tokens as equal.

3. Complete: "We can reduce the allowance, provided that ____."

- A** the summary covering the required points
- B** cover the summary the required points
- C** the summary still covers the required points
- D** does the summary cover the required points

4. Which claim is appropriately limited?

- A** Removing duplicates proves the policy is current.
- B** The request fits; we still need to test summary coverage.
- C** A fitting request must produce a complete answer.
- D** More context eliminates all reasoning errors.

Answers and explanations: [page 87](#).

A larger answer will not make the input fit

E Conversations / turns 1-10

Complete the numbered gaps. Use each bank entry once, without changing its form. Cover the answers at the end until you finish.

WORD + PHRASE BANK

token budget / capacity constraint / position / exception clause / truncation / delimiters / long-context / deduplication / reservation / headroom

- 01 **Lina:** The summary request is rejected again. I was going to increase the output allowance so the model has enough room to explain the policy.
- 02 **Omar:** Can we check the ¹ _____ first? Your input is 29,000 tokens, and you have reserved another 4,000 for output.
- 03 **Lina:** That sounds close to the 32,000 limit. Are you saying the unused output allowance counts even though the model has not answered yet?
- 04 **Omar:** For this configuration, yes. The combined ² _____ is 33,000. Increasing the output allowance would move us further over the limit.
- 05 **Lina:** All right, I had treated the limit as input only. Could we simply trim the final document until the request is accepted?
- 06 **Omar:** Not without checking what it contains. The final pages include an ³ _____ that changes the refund rule for renewals.
- 07 **Lina:** We also repeat the same header on every page. Removing those exact copies would preserve the policy wording and reduce the input somewhat.
- 08 **Omar:** That is a reasonable first test. Use exact ⁴ _____, then count again with the tokenizer used by this model rather than a word count.
- 09 **Lina:** Suppose that saves only 600 tokens. We would still be 400 over the limit. Could we reserve 3,000 output tokens instead?
- 10 **Omar:** We could test it. That would leave 600 tokens of ⁵ _____, but we need to verify that the shorter answer still covers the required points.

The conversation, continued

E Conversations / turns 11-20

Continue with the word bank on page 19.

- 11 **Lina:** I'll check renewals and exceptions against the same checklist. Getting the request accepted won't help if the summary drops the important part.
- 12 **Omar:** Exactly. Also record whether the client performs ⁶ _____. A request can fit after important text has been removed without anyone noticing.
- 13 **Lina:** Someone suggested using a larger-context model instead. Would that let us keep the original request and stop spending time on these smaller adjustments?
- 14 **Omar:** It is another option, not an automatic solution. Its ⁷ _____ performance on our policy task has not been evaluated, and its cost may differ.
- 15 **Lina:** Then I will keep that as a separate comparison. Should the test vary where the exception appears, or is the current document order sufficient?
- 16 **Omar:** Vary its ⁸ _____ while keeping the wording unchanged. We want to see whether the model preserves the exception across the tested arrangements.
- 17 **Lina:** I'll remove the repeated headers first. If we're still over, I'll test the smaller output allowance and bring both the counts and the coverage results.
- 18 **Omar:** Correct. Keep the ⁹ _____ between documents so the revised input does not merge separate policies into one undifferentiated block.
- 19 **Lina:** I will report the before-and-after counts with the coverage results. If the necessary clauses still cannot fit, I will stop rather than silently remove them.
- 20 **Omar:** Good. Describe the limit as a ¹⁰ _____, not a model-quality finding. We can then choose the next workflow using the actual test results.

ANSWERS / EXPLANATIONS: PAGE 88

1. token budget | 2. reservation | 3. exception clause | 4. deduplication | 5. headroom | 6. truncation | 7. long-context | 8. position | 9. delimiters | 10. capacity constraint

A contract summary with repeated appendices

F Say it

A fictional request has a 20,000-token total limit, 18,000 input tokens, and a 3,000-token output reservation. Repeated appendix covers may be removed; unique obligations must remain.

Complete the four gaps. Use each bank entry once.

WORD + PHRASE BANK

obligation / coverage / reservation / deduplication

Engineer: "The combined ¹ _____ is 21,000 tokens, so the request is too large."

Reviewer: "Start with exact ² _____ of the repeated covers."

Engineer: "I will preserve every unique ³ _____ in the supplied contract."

Reviewer: "After recounting, test ⁴ _____ rather than treating acceptance as success."

ANSWERS / EXPLANATIONS: PAGE 88

1. reservation | 2. deduplication | 3. obligation | 4. coverage

Rehearse with a fixed script

1. After checking the dialogue, read the reservation calculation aloud: 29,000 input plus 4,000 reserved output equals 33,000 against a 32,000 limit.
2. Switch roles. Read the proposal to remove exact duplicates while retaining the exception clauses; do not treat a fitting request as proof of coverage.
3. Complete the appendix exchange. Check 18,000 plus 3,000 equals 21,000, then preserve unique obligations while comparing the total with the fictional 20,000 limit.

WORKING ALONE

Read both roles, pausing for two seconds when the speaker changes. Repeat with the bank covered, then check any missed expressions.

A 94 percent score with an overlap problem

A Read the situation

Disclose compromised evidence and resist pressure to overstate readiness.

A classifier scores 94 percent on 500 evaluation examples. Jules finds that 40 evaluation records have exact copies in training. The remaining records have not been re-scored separately. Morgan presents release evidence in one hour. Training and evaluation labels use the same written rubric, but six disputed labels remain unresolved. The release owner can postpone the decision.

PEOPLE / Jules: Data scientist | Morgan: Evaluation lead

REPORT EVIDENCE QUALITY BEFORE HEADLINE PERFORMANCE

Flagging contaminated evaluation data is responsible reporting, not an admission that every model result is worthless. State exactly what is affected, what cannot yet be concluded, and which decision needs a revised evidence base.

Briefing checks

Circle one answer in each item. Use the briefing.

1. What does 94 percent currently describe?

- A** A verified production success rate
- B** The agreement rate on disputed labels
- C** The score on all 500 evaluation examples
- D** The score on 460 nonoverlapping examples

2. What overlap has been confirmed?

- A** Six hundred records are duplicates.
- B** Forty evaluation records have exact training copies.
- C** Every evaluation record is in training.
- D** Forty training labels are missing.

3. What can the release owner do?

- A** Turn the score into a production guarantee
- B** Postpone the decision
- C** Certify the unmeasured subset score
- D** Resolve labels without review

Answers and explanations: [page 89](#).

The vocabulary that does the work

B Find the words / 1 of 2

Read the definitions and say the common word combinations aloud.

dataset

A collection of records assembled for a purpose.

IN USE curate a dataset

training set

Data used to fit model parameters.

IN USE audit the training set

validation set

Data used to guide model selection or tuning.

IN USE monitor validation performance

test set

Data reserved for evaluating a finalized choice.

IN USE preserve the test set

holdout

Data kept separate from a fitting or selection process.

IN USE maintain a clean holdout

data leakage

Information crossing a boundary in a way that invalidates evaluation.

IN USE investigate data leakage

contamination

Unwanted overlap or exposure that weakens an evaluation.

IN USE quantify evaluation contamination

exact duplicate

A record identical under a stated comparison rule.

IN USE remove exact duplicates

near duplicate

A closely similar record that may share substantive content.

IN USE review near duplicates

split

A partition of data into separate subsets.

IN USE define a reproducible split

label

A target category or annotation attached to a record.

IN USE verify a disputed label

annotation

Human or machine-added information about an example.

IN USE review annotation quality

Precision check

A clean holdout is separate from the process it evaluates, not necessarily representative of production. Removing overlap improves separation but does not resolve label disputes or distribution differences.

The vocabulary that does the work

B Find the words / 2 of 2

03

Read the definitions and say the common word combinations aloud.

rubric

Explicit criteria for assigning or evaluating labels.

IN USE apply a shared rubric

adjudication

Resolution of a disputed judgment by a defined process.

IN USE request label adjudication

inter-rater agreement

The degree to which different raters agree.

IN USE measure inter-rater agreement

provenance

The recorded origin and history of data.

IN USE establish data provenance

sampling frame

The population or list from which records are sampled.

IN USE define the sampling frame

representativeness

How well a sample reflects the target population.

IN USE assess sample representativeness

class imbalance

Unequal frequencies among target categories.

IN USE check class imbalance

stratification

Sampling or partitioning within defined groups.

IN USE use stratified sampling

distribution shift

A difference between data distributions across settings.

IN USE detect distribution shift

deduplicate

Remove repeated records under a defined rule.

IN USE deduplicate before splitting

denominator

The total count used as the base of a rate.

IN USE state the denominator

re-score

Calculate a score again under specified conditions.

IN USE re-score the clean subset

Precision check

Exact duplicates and near duplicates require different comparison rules. Inter-rater agreement measures agreement, not automatic correctness. Always identify the sample and denominator behind a percentage.

Phrases you can actually use

C Notice the language / 1 of 2

03

FLAG COMPROMISED EVIDENCE

"The headline score includes records that overlap with training."

PROTECT UNCERTAINTY

"We have not yet calculated the score on the remaining records."

SEPARATE TWO PROBLEMS

"Data overlap and disputed labels need different checks."

NAME THE DECISION IMPACT

"The current score should not be presented as an independent estimate."

QUANTIFY THE ISSUE

"We found 40 exact overlaps among 500 evaluation examples."

REJECT UNSUPPORTED SUBTRACTION

"Removing eight percent of records does not tell us the new accuracy."

REQUEST PROVENANCE

"Can we trace when these records entered each split?"

OFFER A BOUNDED UPDATE

"I can provide the overlap list before the meeting."

Includes versus consists of

"The set includes overlapping records" does not mean every record overlaps.

Not yet

"Not yet re-scored" describes incomplete work; it does not imply the corrected result will improve.

Percentage points

A change from 94 percent to 90 percent is four percentage points, not necessarily a four-percent relative decline.

Phrases you can actually use

C Notice the language / 2 of 2

03

CLARIFY A DUPLICATE

"What comparison rule defines an exact match here?"

QUALIFY A CLEAN RESULT

"A clean split still needs a representativeness check."

STATE THE DENOMINATOR

"Please report both the number correct and the number evaluated."

KEEP THE RECORD

"Retain the original score with its limitation and the corrected result."

PRESERVE THE HOLDOUT

"Do not tune against the replacement test set."

CHECK THE RUBRIC

"Which label criterion produced the disagreement?"

DESCRIBE A LIMITATION

"This sample underrepresents the rare class."

ESCALATE A DECISION

"The release owner needs to decide whether to wait for the revised evidence."

Reported speech

"The slide says the holdout is independent" identifies a claim without endorsing it.

Known and unknown

Pair "we found 40 overlaps" with "we have not checked near duplicates" when both are relevant.

Accountable correction

"We need to correct the slide" directs an action without assigning unsupported personal blame.

Choose the precise message

D Check your understanding

03

Circle one best answer for each item. Use the stated purpose and case facts, not the option that merely sounds most polite.

1. Which meeting statement is supported?

- A** Every label is unreliable.
- B** The model will fail in production.
- C** The 94 percent score includes 40 training overlaps.
- D** The clean-subset score is exactly 86 percent.

2. What does "not yet re-scored" mean?

- A** The revised score must be lower.
- B** The release has been canceled permanently.
- C** The labels are now verified.
- D** The revised calculation remains unfinished.

3. Which question targets a label dispute?

- A** Which rubric criterion distinguishes these two labels?
- B** Can we use the model's own prediction as the reference label?
- C** Does removing duplicates prove the disputed labels are correct?
- D** Can the headline score decide which reviewer is right?

4. Which phrase correctly names the base of a rate?

- A** state the denominator
- B** approve the leakage
- C** deploy the annotation
- D** infer the rubric

Answers and explanations: [page 89](#).

A 94 percent score with an overlap problem

E Conversations / turns 1-10

Complete the numbered gaps. Use each bank entry once, without changing its form. Cover the answers at the end until you finish.

WORD + PHRASE BANK

re-score / representativeness / provenance / model selection / holdout / adjudication / contamination / limitation statement / exact duplicates / denominator

- 01 **Morgan:** The release meeting starts in an hour. I have the 94 percent result on the slide. Is there anything material we need to add?
- 02 **Jules:** Yes. I found 40 exact training matches inside the 500-example evaluation set. That creates ¹ _____, so the independence claim needs correcting.
- 03 **Morgan:** Forty out of 500 is eight percent. Could I subtract eight points and present 86 percent as a conservative estimate instead?
- 04 **Jules:** No. We need to ² _____ the remaining examples. We do not know how many of the overlapping records were answered correctly.
- 05 **Morgan:** Then I will remove the independent-evaluation label. Can you tell me whether the overlap came from the import process or the later cleanup?
- 06 **Jules:** Not yet. The ³ _____ log should help, but I have not reconstructed the sequence. I can provide the exact record list now.
- 07 **Morgan:** Send me that list. I'll leave the cause out of the slide for now. Is there anything else I need to flag?
- 08 **Jules:** Six examples have disputed labels. They need ⁴ _____ under the written rubric, independently of the overlap check.
- 09 **Morgan:** Could we exclude those six for this meeting? I do not want the discussion to get buried in detail that may not change the decision.
- 10 **Jules:** We can report an explicitly defined subset, but we must state its ⁵ _____ and exclusions. We cannot call a selectively cleaned subset the original result.

The conversation, continued

E Conversations / turns 11-20

03

Continue with the word bank on page 28.

- 11 **Morgan:** Agreed. The release owner can wait for better evidence. What would you recommend we say about the 94 percent figure already circulated yesterday?
- 12 **Jules:** Retain it with a correction: it describes the original set, not a clean
6 _____. Replace the readiness claim rather than quietly changing the number.
- 13 **Morgan:** Once the duplicate records are removed, can we say the evaluation represents customer traffic? That is the next question I expect from the release owner.
- 14 **Jules:** Not from separation alone. We still need to check 7 _____, including the rare category that appears only a few times in this sample.
- 15 **Morgan:** So we have three issues: overlap, disputed labels, and coverage of the target population. Which one can you resolve before today's meeting starts?
- 16 **Jules:** I can verify the 8 _____ and document the matching rule. I cannot responsibly promise all label reviews within the hour.
- 17 **Morgan:** All right, the duplicate check comes first. For the replacement set, can we keep it untouched while the team tunes the model?
- 18 **Jules:** It should. If we repeatedly choose model settings from that set, it becomes part of
9 _____, and we need a separate final test.
- 19 **Morgan:** I will recommend postponing the readiness decision. The slide will list confirmed overlap, pending label review, and the missing clean-subset score separately.
- 20 **Jules:** I will send the records and a short 10 _____. That gives the owner usable evidence without pretending the corrected outcome is already known.

ANSWERS / EXPLANATIONS: PAGE 90

1. contamination | 2. re-score | 3. provenance | 4. adjudication | 5. denominator | 6. holdout | 7. representativeness | 8. exact duplicates | 9. model selection | 10. limitation statement

A sentiment test with repeated customer messages

F Say it

An 80-message test contains ten copies of messages used for training. The team has not scored the 70 remaining messages separately. Two reviewers disagree on three labels.

Complete the four gaps. Use each bank entry once.

WORD + PHRASE BANK

overlap / denominator / re-score / adjudication

Analyst: "The current test has training ¹ _____, so its independence is compromised."

Lead: "Report the clean-subset score only after you ² _____ those 70 messages."

Analyst: "The three disputed labels also need ³ _____."

Lead: "And include the ⁴ _____ with every reported percentage."

ANSWERS / EXPLANATIONS: PAGE 90

1. overlap | 2. re-score | 3. adjudication | 4. denominator

Rehearse with a fixed script

1. Check the answers. Read 40 overlaps out of 500 as eight percent of records, not eight percentage points to subtract from the 94 percent score.

2. Switch roles. Separate the overlap list, unmeasured clean-subset score and six disputed labels in the handoff.

3. Complete the 80-message exchange. Retain 70 unscored nonoverlapping messages and three label disputes; do not invent a revised accuracy.

WORKING ALONE

Read both roles, pausing for two seconds when the speaker changes. Repeat with the bank covered, then check any missed expressions.

More relevant passages, a slower response

A Read the situation

04

Negotiate a trade-off using matched comparisons and explicitly limited claims.

A document assistant with reranking retrieves a relevant passage for 86 of 100 fixed test queries, compared with 78 without reranking. The reranker adds 300 milliseconds to the measured retrieval path. The generator and query set are unchanged. No user study or end-to-end answer-quality review has been completed. Sam recommends a pilot; Elena owns the evaluation plan.

PEOPLE / Sam: Retrieval engineer | Elena: Evaluation lead

DEFINE BETTER BEFORE AGREEING

"Better retrieval" may mean finding relevant passages, ranking them earlier, or improving final answers. Ask which measure changed. A measured technical improvement can justify another test without proving a better customer experience.

Briefing checks

Circle one answer in each item. Use the briefing.

1. What improved in the fixed test?

A Queries with a relevant retrieved passage rose from 78 to 86.

C Response time fell by 300 milliseconds.

B Every final answer became correct.

D Users preferred the new experience.

2. Which component stayed unchanged?

A The reranker

C The number of relevant hits

B The retrieval path timing

D The generator

3. Which claim needs further evidence?

A The reranker adds 300 milliseconds.

C The test contains 100 queries.

B Both variants use the same query set.

D The new path improves end-to-end answer quality.

Answers and explanations: page 91.

The vocabulary that does the work

B Find the words / 1 of 2

Read the definitions and say the common word combinations aloud.

RAG

Retrieval-augmented generation; generating with retrieved context.

IN USE evaluate a RAG pipeline

embedding

A numerical representation used to compare or process items.

IN USE compute an embedding

vector

An ordered collection of numerical values.

IN USE compare document vectors

vector store

A store or index supporting retrieval of vector representations.

IN USE query a vector store

similarity search

Finding items close under a chosen similarity measure.

IN USE run similarity search

semantic search

Retrieval based on meaning-related representations.

IN USE assess semantic search

keyword search

Retrieval based on words or lexical matches.

IN USE combine keyword search with vectors

hybrid retrieval

Retrieval combining different search approaches.

IN USE test hybrid retrieval

chunk

A retrievable segment of a larger document.

IN USE retrieve a relevant chunk

chunk boundary

The point where one segment ends and another starts.

IN USE preserve context at chunk boundaries

overlap window

Content shared between adjacent chunks.

IN USE adjust the overlap window

reranker

A component that reorders candidates for relevance.

IN USE evaluate the reranker

Precision check

Retrieval relevance, answer correctness, and user preference are distinct outcomes. A citation can identify a source without proving that the source supports the exact claim. Inspect the cited passage.

The vocabulary that does the work

B Find the words / 2 of 2

Read the definitions and say the common word combinations aloud.

candidate set

Items available for a later selection or ranking step.

IN USE inspect the candidate set

top-k

The first k results returned by a ranking.

IN USE compare top-k results

relevance

How well material addresses a specified query.

IN USE judge passage relevance

recall

The proportion of relevant items retrieved under a defined evaluation.

IN USE measure retrieval recall

precision

The proportion of retrieved items judged relevant.

IN USE improve retrieval precision

grounding

Connecting generated claims to supporting information.

IN USE check answer grounding

citation

A reference identifying a supporting source.

IN USE verify a citation

source attribution

Identification of where information came from.

IN USE preserve source attribution

index freshness

How current the indexed information is.

IN USE monitor index freshness

filter

A rule limiting eligible retrieval results.

IN USE apply a metadata filter

ablation

A comparison that removes or changes a component.

IN USE run a reranker ablation

end-to-end

Covering the complete path from request to outcome.

IN USE measure end-to-end latency

Precision check

Recall and precision require defined relevance judgments and denominators. The case reports query-level passage coverage, not a complete precision-recall analysis. Top-k says how many results are returned, not how many are correct.

Phrases you can actually use

C Notice the language / 1 of 2

NAME THE MEASURE

"The improvement is passage coverage on this fixed query set."

QUANTIFY A TRADE-OFF

"The gain comes with 300 milliseconds of added retrieval time."

ASK FOR AN ABLATION

"What happens when we remove only the reranker?"

CHECK THE SOURCE

"Does the cited passage support this specific sentence?"

SEPARATE OUTCOMES

"We have not yet measured final-answer accuracy."

REQUEST A MATCHED COMPARISON

"Can we hold the generator and queries constant?"

LIMIT A RECOMMENDATION

"These results support a bounded pilot, not a general rollout claim."

DEFINE THE NEXT TEST

"Review final answers and timing on the same cases."

Compared with

State both variants: "86 queries compared with 78 without reranking." The comparison is then recoverable.

Percentage points

On this 100-query set, coverage rises by eight percentage points, not merely "eight percent better."

Not yet measured

This phrase marks missing evidence without predicting what the eventual test will show.

Phrases you can actually use

C Notice the language / 2 of 2

INSPECT A MISS

"Was the relevant passage absent or merely ranked too low?"

PRESERVE CONTEXT

"The exception and the main rule should not be separated without context."

DISTINGUISH FRESHNESS

"A relevant passage may still come from an outdated index."

COMPARE ALTERNATIVES

"Let us compare reranking all queries with a defined routing policy."

CHECK ELIGIBILITY

"Did a filter exclude the document before similarity search?"

CHALLENGE AN AVERAGE

"Which query types account for most of the added time?"

AVOID CAUSAL OVERREACH

"The matched test isolates this change, not every production condition."

RECORD A LIMITATION

"User preference remains unmeasured in this experiment."

With versus because

"With reranking" describes a condition; "because of reranking" makes a causal claim that needs controlled evidence.

Conditional recommendations

"I recommend a pilot provided we review answer quality" attaches the condition to the action.

Question precision

"Which query types slow down?" is more diagnostic than "Is performance okay?"

Choose the precise message

D Check your understanding

Circle one best answer for each item. Use the stated purpose and case facts, not the option that merely sounds most polite.

1. Which comparison correctly states the observed gain?

- A** Every query returns eight more passages.
- B** Coverage increased by eight percentage points on the fixed set.
- C** Users became eight percent more satisfied.
- D** Final answers are now 86 percent accurate.

2. What is a reranker ablation here?

- A** Remove all supporting documents permanently.
- B** Compare the path with and without the reranker while holding other factors fixed.
- C** Replace every component and change the queries.
- D** Ask users to invent new relevance criteria.

3. Which sentence preserves the limitation?

- A** Better retrieval guarantees better answers.
- B** A citation proves the answer is true.
- C** No user study means retrieval did not improve.
- D** Retrieval improved on this set; final-answer quality remains unmeasured.

4. Which question checks grounding?

- A** Is the cited document present in the current index?
- B** Does the retrieved passage share many words with the question?
- C** Does this passage support the answer's refund exception?
- D** Does the answer contain a citation marker in the required format?

Answers and explanations: [page 91](#).

More relevant passages, a slower response

E Conversations / turns 1-10

04

Complete the numbered gaps. Use each bank entry once, without changing its form. Cover the answers at the end until you finish.

WORD + PHRASE BANK

index freshness / query type / trade-off / conditional / end-to-end / reranker / grounding / chunk boundary / candidate set / ablation

- 01 **Sam:** The reranker looks promising. We found a relevant passage for 86 of the same 100 queries, compared with 78 on the previous path.
- 02 **Elena:** Good. Before we call it better overall, what changed in the
1 _____? Did you also modify the initial search or document filters?
- 03 **Sam:** No. The initial candidates, generator, and queries are unchanged. The reranker changes their order, and the retrieval path takes another 300 milliseconds.
- 04 **Elena:** Then we have a useful 2 _____. It shows a retrieval gain with added time, but it does not yet show better generated answers.
- 05 **Sam:** I agree the answer review is unfinished. My recommendation is a small pilot, not a rollout. Which evidence would you need before approving that pilot?
- 06 **Elena:** First, an 3 _____ review of the same cases. We need the final answers, their supporting passages, and the complete response times.
- 07 **Sam:** We can collect those. Some of the original misses had the right document in the candidates but ranked below the five passages sent to generation.
- 08 **Elena:** That distinction matters. A 4 _____ can promote those passages, but it cannot rescue a relevant document that never entered the candidate set.
- 09 **Sam:** Two misses also involve a policy exception split across adjacent chunks. I do not want to alter chunking in the middle of this comparison.
- 10 **Elena:** Keep that separate. Log the 5 _____ issue for another experiment so we can still interpret this result.

The conversation, continued

E Conversations / turns 11-20

04

Continue with the word bank on page 37.

- 11 **Sam:** Should the answer reviewers count every displayed citation as a success? The new path tends to produce more citations, which looks reassuring in the interface.
- 12 **Elena:** No. They should check ⁶ _____ for each substantive claim. A citation can point to a real document that does not support that sentence.
- 13 **Sam:** Understood. I will include the exact cited passage and the answer claim together. We also need to distinguish an old policy from an irrelevant one.
- 14 **Elena:** Yes. Check ⁷ _____ separately. A highly relevant passage from a superseded policy can still produce a misleading answer.
- 15 **Sam:** The product lead will ask whether customers will accept the extra delay. Could we infer that from the eight-point increase in passage coverage?
- 16 **Elena:** No. That is a ⁸ _____ we have measured technically, but customer preference remains unknown until we test it appropriately.
- 17 **Sam:** Then I will present the retrieval result as eight percentage points on this set. I will not call it an eight-percent increase in customer satisfaction.
- 18 **Elena:** Exactly. Also break down timing by ⁹ _____, because the same average can hide very different costs for simple and complex questions.
- 19 **Sam:** I'll bring the answer review and timing breakdown before we discuss the pilot. The chunking experiment can wait; I don't want to lose this comparison.
- 20 **Elena:** Good. Put the improvement on the slide, but keep the pilot recommendation ¹⁰ _____. I want to see the actual answers before I approve it.

ANSWERS / EXPLANATIONS: PAGE 92

1. candidate set | 2. ablation | 3. end-to-end | 4. reranker | 5. chunk boundary | 6. grounding | 7. index freshness | 8. trade-off | 9. query type | 10. conditional

An answer cites the right document but the wrong section

04

F Say it

A benefits assistant cites the current policy but uses the standard rule instead of a stated exception. Retrieval includes both sections. A reviewer must report claim support, not count citation markers.

Complete the four gaps. Use each bank entry once.

WORD + PHRASE BANK

grounding / context / end-to-end / source

Reviewer: "The citation identifies a real ¹ _____, but not support for this claim."

Engineer: "Both sections were retrieved, so we should inspect answer ² _____."

Reviewer: "The exception is missing from the answer, although it is present in the ³ _____."

Engineer: "We will report that distinction in the ⁴ _____ review."

ANSWERS / EXPLANATIONS: PAGE 92

1. source | 2. grounding | 3. context | 4. end-to-end

Rehearse with a fixed script

1. Check the dialogue. Read 78 versus 86 of 100: eight percentage points gained, with 300 milliseconds added.
2. Switch roles. Request a matched final-answer review; do not substitute retrieval coverage for answer accuracy.
3. Complete the exception exchange. Distinguish a real source from support for the specific claim.

WORKING ALONE

Read both roles, pausing for two seconds when the speaker changes. Repeat with the bank covered, then check any missed expressions.

One proposal, two different failure modes

A Read the situation

Separate proposed remedies by mechanism and agree on a controlled experiment.

A support summarizer sometimes returns paragraphs instead of the required fields and sometimes cites an obsolete refund policy. The obsolete wording is present in its retrieval index. Its prompt contains only one formatting example. Tessa proposes fine-tuning to solve both problems. Dev owns the experiment plan. The team has 200 permission-cleared example summaries but has not audited their label consistency.

PEOPLE / Tessa: Applied scientist | Dev: Technical lead

CHALLENGE THE SCOPE OF A REMEDY

A method can be useful without solving every problem in the discussion. Separate knowledge freshness from output behavior. Ask what mechanism would change each failure and what simpler comparison could test the claim.

Briefing checks

Circle one answer in each item. Use the briefing.

1. Which fact directly concerns outdated information?

- A The retrieval index contains obsolete policy wording.
- C The team owns an experiment plan.

- B The prompt contains one formatting example.
- D The output sometimes uses paragraphs.

2. What is known about the 200 examples?

- A Every label has been independently verified.
- C They guarantee successful fine-tuning.

- B They contain the latest policy by definition.
- D They are permission-cleared but consistency is unaudited.

3. What distinction should the experiment preserve?

- A Data-use permission as proof of label consistency
- C More training examples as proof that retrieval is refreshed

- B Required fields as proof that policy statements are current
- D Source freshness versus output-format behavior

Answers and explanations: [page 93](#).

The vocabulary that does the work

B Find the words / 1 of 2

Read the definitions and say the common word combinations aloud.

fine-tuning

Updating model parameters using additional training data.

IN USE evaluate a fine-tuning run

pretraining

Broad initial training before task-specific adaptation.

IN USE distinguish pretraining from adaptation

supervised fine-tuning

Training on supplied input-output examples.

IN USE prepare supervised fine-tuning data

SFT

Abbreviation for supervised fine-tuning.

IN USE compare SFT with prompting

preference data

Examples recording preferences between candidate outputs.

IN USE audit preference data

alignment

Shaping behavior toward specified goals or preferences.

IN USE define the alignment objective

RLHF

Reinforcement learning from human feedback.

IN USE discuss an RLHF workflow

DPO

Direct preference optimization; a preference-training method.

IN USE evaluate a DPO experiment

LoRA

Low-rank adaptation; a parameter-efficient adaptation method.

IN USE train a LoRA adapter

adapter

A small trainable component used for adaptation.

IN USE load a task adapter

base model

The starting model before the adaptation under discussion.

IN USE compare against the base model

training objective

The quantity a training process seeks to optimize.

IN USE specify the training objective

Precision check

Fine-tuning changes model parameters; prompting supplies instructions at inference time; retrieval supplies external context. These interventions can interact, but they do not have interchangeable mechanisms or guarantees.

The vocabulary that does the work

B Find the words / 2 of 2

Read the definitions and say the common word combinations aloud.

hyperparameter

A training or model setting selected rather than learned in that run.

IN USE document a hyperparameter choice

learning rate

A setting controlling the scale of training updates.

IN USE compare learning-rate settings

epoch

One pass through the training data.

IN USE record the epoch count

overfitting

Fitting training patterns without adequate generalization.

IN USE monitor for overfitting

generalization

Performance on relevant examples beyond those used for fitting.

IN USE test generalization

catastrophic forgetting

Loss of previously learned capabilities during further training.

IN USE assess catastrophic forgetting

format adherence

Consistency with a required output structure.

IN USE measure format adherence

structured output

Output constrained or organized into a defined structure.

IN USE validate structured output

few-shot prompting

Supplying a small set of examples in the prompt.

IN USE test few-shot prompting

behavioral target

The specified behavior an intervention aims to change.

IN USE define the behavioral target

training artifact

A saved output of training, such as weights or configuration.

IN USE version the training artifact

rollback candidate

A previously validated configuration available for reversion.

IN USE retain a rollback candidate

Precision check

Permission-cleared examples are authorized for use, not automatically accurate or consistently labeled. Format adherence is a measurable behavioral target; it does not prove policy correctness or broad alignment.

Phrases you can actually use

C Notice the language / 1 of 2

SEPARATE FAILURES

"We have a formatting issue and a source-freshness issue."

DEFINE A TARGET

"The target is consistent output in the required fields."

CHECK EXAMPLES

"Are the reference summaries consistent with the same rubric?"

HOLD A VARIABLE FIXED

"Use the same cleaned source set for both configurations."

ASK ABOUT MECHANISM

"How would this change prevent the obsolete policy from entering the context?"

PROPOSE A BASELINE

"Let us test clearer prompting before attributing gains to training."

LIMIT A REMEDY

"Fine-tuning may improve the behavior; it does not keep an index current."

AGREE ON EVIDENCE

"We will compare format adherence and factual correctness separately."

May, not will

"Fine-tuning may improve adherence" expresses a hypothesis; "will fix it" promises an untested result.

Two-object contrast

"Change the source, not just the behavior" draws attention to what each intervention affects.

Before attributing

This phrase delays a causal conclusion until a comparison has been completed.

Phrases you can actually use

C Notice the language / 2 of 2

PROTECT EVALUATION

"Keep the final test examples outside the training set."

ASK FOR PERMISSION STATUS

"Are these examples cleared for the intended training use?"

CHECK WIDER EFFECTS

"Does the adaptation reduce performance on other required tasks?"

BOUND A RECOMMENDATION

"I recommend an experiment, not immediate adoption."

STATE AN ALTERNATIVE

"A constrained output format may address part of this problem."

AVOID CONFLATION

"Permission, quality, and representativeness are separate checks."

KEEP A BASELINE

"Retain the validated base configuration for comparison."

RECORD THE DECISION

"We will fix the stale index and test the formatting intervention independently."

Separate adverbs

"Evaluate them separately" refers to separate measures, not necessarily separate teams.

What would

"What would count as success?" requests a criterion without presupposing success.

Provided that

Use the condition in the same sentence: "We can train, provided that the data audit passes."

Choose the precise message

D Check your understanding

Circle one best answer for each item. Use the stated purpose and case facts, not the option that merely sounds most polite.

1. Which action directly addresses the known stale source?

- A** Assume format training refreshes all documents.
- B** Rename the output fields.
- C** Increase the number of training epochs blindly.
- D** Correct the retrieval index through the approved update process.

2. Which is a behavioral target?

- A** Owning a permission-cleared file
- B** Scheduling a review meeting
- C** Having a larger office
- D** Returning the required fields consistently

3. Which claim needs qualification?

- A** Fine-tuning will solve both failures.
- B** The prompt has one example.
- C** The index contains obsolete wording.
- D** The label audit is unfinished.

4. Which comparison best isolates the formatting intervention?

- A** Evaluate only the training examples.
- B** Accept any output with a confident tone.
- C** Keep sources and test cases fixed while changing the formatting method.
- D** Change data, model, metrics, and test cases together.

Answers and explanations: [page 93](#).

One proposal, two different failure modes

E Conversations / turns 1-10

05

Complete the numbered gaps. Use each bank entry once, without changing its form. Cover the answers at the end until you finish.

WORD + PHRASE BANK

hypotheses / schema / consistency / catastrophic forgetting / rollback candidate / generalization / format adherence / failure modes / few-shot / behavioral target

- 01 **Tessa:** I think we should fine-tune the summarizer. It ignores our field structure and sometimes quotes the old refund policy. Training could make it more consistent.
- 02 **Dev:** Consistency is worth testing, but those are different ¹ _____. What would the training change in the source material supplied to the model?
- 03 **Tessa:** It would not update the retrieval index. I checked this morning, and the obsolete refund paragraph is still there. That source needs a separate correction.
- 04 **Dev:** Right. Let us define the formatting problem as a ² _____, then keep policy freshness as a data-maintenance issue with its own owner.
- 05 **Tessa:** For formatting, we have 200 example summaries cleared for training use. I can start preparing them, although we have not audited their labels yet.
- 06 **Dev:** Please do that audit first. Permission to use an example does not establish its ³ _____ with the field schema or the instructions in the other examples.
- 07 **Tessa:** Would you want a prompting comparison before training? The current prompt includes only one example, and it does not show how to handle an absent field.
- 08 **Dev:** Yes. A stronger ⁴ _____ baseline and a structured-output option would tell us whether training adds value beyond clearer task specification.
- 09 **Tessa:** All right. Two work items, then: repair the index and compare formatting methods. I'll keep their results separate.
- 10 **Dev:** Exactly. Keep the source set fixed within the comparison, and measure ⁵ _____ separately from policy correctness. An answer can pass one and fail the other.

The conversation, continued

E Conversations / turns 11-20

Continue with the word bank on page 46.

- 11 **Tessa:** Should the training objective reward the presence of every field, even if the source gives no evidence for a value? That could encourage invented details.
- 12 **Dev:** No. The examples need an agreed representation for missing information. A complete
6 _____ does not mean every field contains a positive factual assertion.
- 13 **Tessa:** I will include that distinction in the rubric. After we select the method, we should evaluate on cases that were never used to tune it.
- 14 **Dev:** Correct. We need evidence of 7 _____, including summaries with exceptions, missing details, and layouts absent from the training examples.
- 15 **Tessa:** If we test an adapter, I also want to check our other summary tasks. Better performance on refunds could conceal a regression in cancellation summaries.
- 16 **Dev:** Include those. Further training can affect existing behavior, including
8 _____. We should assess relevant capabilities rather than assuming a small adaptation has no wider effects.
- 17 **Tessa:** I will version the examples, configuration, and resulting adapter together. That way the team can reproduce the comparison and identify exactly what was evaluated.
- 18 **Dev:** Good. Retain a validated 9 _____ as well. A successful experiment is not a reason to lose the previous working configuration.
- 19 **Tessa:** I'll start with the example audit and the prompting baseline. Can we review those results before we spend time on an adapter?
- 20 **Dev:** Yes. List the expected gains as 10 _____ in the experiment note. If the baseline solves the problem, we may not need training.

ANSWERS / EXPLANATIONS: PAGE 94

1. failure modes | 2. behavioral target | 3. consistency | 4. few-shot | 5. format adherence | 6. schema | 7. generalization | 8. catastrophic forgetting | 9. rollback candidate | 10. hypotheses

A classifier uses an outdated category list

F Say it

A classifier is required to use five current categories. Its prompt still lists six older categories. The team has not yet tested a corrected prompt or new training data.

Complete the four gaps. Use each bank entry once.

WORD + PHRASE BANK

hypothesis / baseline / held-out / category list

Scientist: "The prompt supplies an obsolete ¹ _____, so we should correct that input first."

Lead: "Then compare the corrected prompt with the existing ² _____."

Scientist: "We will measure adherence on ³ _____ examples, not only the demonstrations."

Lead: "Until then, the benefit of further training remains a ⁴ _____."

ANSWERS / EXPLANATIONS: PAGE 94

1. category list | 2. baseline | 3. held-out | 4. hypothesis

Rehearse with a fixed script

1. After checking the answers, identify the two failures aloud: obsolete source wording and inconsistent output fields.
2. Switch roles. Read the proposed experiment with fixed sources and test cases; keep data-use permission separate from the unfinished label audit.
3. Complete the category-list exchange. Correct the six-category prompt to the stated five-category requirement, while leaving any training benefit unconfirmed.

WORKING ALONE

Read both roles, pausing for two seconds when the speaker changes. Repeat with the bank covered, then check any missed expressions.

The average is down, but one slice improved

A Read the situation

Present disaggregated results and challenge an oversimplified release argument.

06

On the same 100-case test, the previous model passes 90 cases and the candidate passes 88. Among 20 long-document cases, passes rise from 12 to 18. Among 80 short-request cases, passes fall from 78 to 70. The scoring rubric is unchanged. Production traffic proportions and the seriousness of individual failures have not been analyzed. Noor chairs the review; Ben prepared the results.

PEOPLE / Noor: Evaluation manager | Ben: Machine-learning engineer

DO NOT HIDE THE INCONVENIENT SLICE

A favorable subgroup does not erase an overall decline, and an overall decline does not erase a useful subgroup improvement. Present both. A defensible recommendation explains what is being traded and which missing evidence could change the decision.

Briefing checks

Circle one answer in each item. Use the briefing.

1. What happens to total passes?

- | | |
|-----------------------------------|---|
| A They stay at 90. | B They rise from 12 to 18 overall. |
| C They fall from 90 to 88. | D They rise from 88 to 90. |

2. Which slice improves?

- | | |
|-----------------------------------|---------------------------------|
| A Long-document cases | B Short-request cases |
| C Every production request | D All failure severities |

3. What remains unknown?

- | | |
|---|---------------------------------------|
| A Whether the rubric changed | B The candidate's total passes |
| C The production mixture and seriousness of failures | D The test-set size |

Answers and explanations: page 95.

The vocabulary that does the work

B Find the words / 1 of 2

Read the definitions and say the common word combinations aloud.

evaluation

A structured assessment against specified criteria.

IN USE design an evaluation

benchmark

A reference test used for comparison.

IN USE interpret a benchmark result

baseline

The reference system or result for a comparison.

IN USE establish a baseline

candidate

The version under consideration.

IN USE evaluate the candidate

regression

A deterioration in previously measured behavior.

IN USE investigate a regression

test case

A defined input and expected criterion or outcome.

IN USE inspect a failed test case

golden set

Curated cases used to check important behavior repeatedly.

IN USE maintain a golden set

pass rate

The fraction of evaluated cases meeting a criterion.

IN USE report the pass rate

acceptance criterion

A stated condition required for acceptance.

IN USE agree on acceptance criteria

slice

A defined subgroup within evaluation data.

IN USE analyze a performance slice

aggregate

A combined measure across multiple cases or groups.

IN USE report the aggregate score

disaggregation

Breaking a combined result into meaningful groups.

IN USE disaggregate by task type

06

Precision check

The aggregate falls by two percentage points. Long-document performance rises from 60 to 90 percent; short-request performance falls from 97.5 to 87.5 percent. These are test-set rates, not verified production estimates.

The vocabulary that does the work

B Find the words / 2 of 2

Read the definitions and say the common word combinations aloud.

severity

The seriousness of a failure's consequences.

IN USE assess failure severity

false positive

A positive decision when the reference outcome is negative.

IN USE inspect false positives

false negative

A negative decision when the reference outcome is positive.

IN USE review false negatives

confusion matrix

A table comparing predicted and reference categories.

IN USE examine the confusion matrix

calibration

Agreement between stated confidence and observed outcomes.

IN USE assess confidence calibration

confidence interval

A range expressing uncertainty under stated statistical assumptions.

IN USE report a confidence interval

statistical significance

Evidence against a null model under a specified statistical test.

IN USE distinguish significance from usefulness

practical significance

The importance of a difference for the actual task.

IN USE assess practical significance

paired comparison

Comparison of results on corresponding cases.

IN USE run a paired comparison

LLM-as-judge

Use of a language model to evaluate outputs.

IN USE calibrate an LLM-as-judge setup

human review

Assessment by people using defined criteria.

IN USE conduct a blinded human review

release gate

A defined decision checkpoint before release.

IN USE satisfy the release gate

Precision check

Statistical significance is not practical importance, and an interval is not a guarantee. A pass rate treats cases according to its stated counting rule; it may conceal unequal consequences among failures.

Phrases you can actually use

C Notice the language / 1 of 2

LEAD WITH BOTH RESULTS

"Overall passes fell, while the long-document slice improved."

DESCRIBE THE REGRESSION

"Short-request passes fell from 78 to 70 out of 80."

SEPARATE IMPACT

"A count alone does not tell us how serious those failures are."

LIMIT A CONCLUSION

"We cannot infer production performance from these counts alone."

SHOW THE BASE

"Eighteen of 20 long-document cases passed."

REQUEST PAIRED CASES

"Which previously passing cases now fail?"

CHECK RELEVANCE

"Does this test mixture resemble the traffic we plan to serve?"

PROPOSE THE NEXT REVIEW

"Compare the changed cases and classify their failure severity."

06

While for contrast

"The total fell while one slice improved" presents simultaneous but contrasting findings.

Out of

"Eighteen out of twenty" keeps the numerator and denominator audible.

Previously and now

These words make a regression concrete: "previously passed, now fails."

Phrases you can actually use

C Notice the language / 2 of 2

CHALLENGE A HEADLINE

"The improved slice should not replace the overall result on the slide."

PRESERVE UNCERTAINTY

"We have not established whether this difference is stable across samples."

AVOID METRIC SHOPPING

"Let us agree on the criteria before selecting the winning score."

OFFER A BOUNDED OPTION

"Routing long documents separately is a proposal to test, not an approved fix."

CHECK CONSISTENCY

"Were both versions scored against the same rubric?"

NAME A GATE

"Which acceptance criterion applies to the short-request regression?"

CHECK A JUDGE

"How closely does the automated judge match the reviewed labels?"

CLOSE THE REVIEW

"We will defer the decision until the changed cases have been assessed."

Cannot infer

This limits a conclusion without claiming that the proposed conclusion must be false.

Would need

"We would need a routing evaluation" names a requirement before adopting a proposed alternative.

Comparatives

Specify the object: "better on long documents" is narrower than "a better model."

Choose the precise message

D Check your understanding

Circle one best answer for each item. Use the stated purpose and case facts, not the option that merely sounds most polite.

1. Which summary preserves both findings?

- A** Every task improved by six points.
- B** Short requests improved from 70 to 78.
- C** Production accuracy is exactly 88 percent.
- D** Total passes fell by two; long-document passes rose by six.

2. What is the candidate's long-document pass rate?

- A** 90 percent
- B** 18 percent
- C** 88 percent
- D** 60 percent

3. Which question targets practical significance?

- A** Do the slice counts add up to the aggregate score?
- B** Did both variants use the same scoring rubric?
- C** How serious are the newly failing cases for users?
- D** Is the two-point difference stable across repeated samples?

4. Which proposed conclusion exceeds the evidence?

- A** The candidate passes 88 of 100 cases.
- B** The short-request slice worsened.
- C** Long-document routing is already safe to release.
- D** The rubric is unchanged.

Answers and explanations: [page 95](#).

The average is down, but one slice improved

E Conversations / turns 1-10

Complete the numbered gaps. Use each bank entry once, without changing its form. Cover the answers at the end until you finish.

06

WORD + PHRASE BANK

statistical significance / severity / regression / release gate / disaggregation / human review / acceptance criteria / false positives / aggregate / paired comparison

- 01 **Ben:** The candidate is much stronger on long documents. It passed 18 of those 20 cases, compared with 12 for the previous version. That seems release-worthy.
- 02 **Noor:** Show the ¹ _____ as well. The candidate passes 88 out of 100 overall, while the previous version passes 90.
- 03 **Ben:** Yes. The short requests account for the decline: 70 passes out of 80, compared with 78. I should have led with both results.
- 04 **Noor:** Thank you. That is a meaningful ² _____ to investigate, even though the long-document gain is real on this set.
- 05 **Ben:** Could we weight the long documents more heavily? They are harder, and the team invested most of its work in that capability during this iteration.
- 06 **Noor:** Difficulty alone does not determine the weight. We need the intended traffic mixture and agreed ³ _____, not weights chosen after seeing the result.
- 07 **Ben:** The production proportions have not been analyzed. We can say that plainly. I can also list which short-request cases changed from pass to fail.
- 08 **Noor:** Please do a ⁴ _____. The totals do not tell us whether the new failures are harmless omissions or serious errors.
- 09 **Ben:** Would you want a severity category for every changed case? A refusal on a valid request and a confident false answer both count as failures now.
- 10 **Noor:** Exactly. The rubric can preserve its binary score while a separate ⁵ _____ review explains the impact. We should not silently change the scoring rule halfway through.

The conversation, continued

E Conversations / turns 11-20

Continue with the word bank on page 55.

- 11 **Ben:** We used an automated judge for the first pass. It agreed with the reviewed labels on most examples, but I have not checked the newly failing slice.
- 12 **Noor:** Then check that slice with ⁶ _____. Agreement across the whole set can hide weaker judgment on the cases that matter here.
- 13 **Ben:** If those reviews confirm the long-document gain, could we route only long documents to the candidate and keep short requests on the previous version?
- 14 **Noor:** It is a useful proposal, but the routing decision becomes another component to evaluate. We have not measured its ⁷ _____ or the consequences of misclassification.
- 15 **Ben:** So routing needs its own test. And twenty long documents isn't much evidence. Should I add an uncertainty estimate to the slide?
- 16 **Noor:** Yes, with appropriate assumptions and qualified analysis. Do not use ⁸ _____ as a synonym for business value or a guarantee of future performance.
- 17 **Ben:** I will revise the slide to show both counts, the shared rubric, and the missing traffic and severity analysis. The candidate's strengths will still be visible.
- 18 **Noor:** Good. That is ⁹ _____, not an attempt to hide the average. We want the decision-maker to see why the headline alone is insufficient.
- 19 **Ben:** I'll recommend waiting for the changed-case review. I still think routing is worth testing, but I won't present it as a fix today.
- 20 **Noor:** Agreed. The ¹⁰ _____ remains open. The next update should say which criteria are met, which are not, and what evidence is still missing.

06

ANSWERS / EXPLANATIONS: PAGE 96

1. aggregate | 2. regression | 3. acceptance criteria | 4. paired comparison | 5. severity | 6. human review | 7. false positives | 8. statistical significance | 9. disaggregation | 10. release gate

A support classifier misses urgent messages

F Say it

A classifier has a high overall score but misses four of ten urgent messages. Urgent-message consequences have not been reviewed. The team must not treat aggregate success as sufficient evidence for release.

06

Complete the four gaps. Use each bank entry once.

WORD + PHRASE BANK

severity / release gate / false negatives / aggregate

Reviewer: "The overall ¹ _____ hides a weak urgent-message subgroup."

Engineer: "Those missed urgent messages are ² _____ for that category."

Reviewer: "We need a separate assessment of their ³ _____."

Engineer: "Then the ⁴ _____ should remain unresolved until that review is complete."

ANSWERS / EXPLANATIONS: PAGE 96

1. aggregate | 2. false negatives | 3. severity | 4. release gate

Rehearse with a fixed script

1. Check the dialogue. Read both slices: 12 to 18 of 20 long cases; 78 to 70 of 80 short cases.
2. Switch roles. State that total passes fall from 90 to 88; severity and production mixture remain unmeasured.
3. Complete the urgent-message exchange. Name the four misses as false negatives and retain the required severity review.

WORKING ALONE

Read both roles, pausing for two seconds when the speaker changes. Repeat with the bank covered, then check any missed expressions.

Faster on average is not ready for everyone

A Read the situation

Negotiate a staged deployment with explicit conditions and rollback ownership.

A smaller model reduces mean response time from four seconds to two on a fixed test. Complex-answer quality and peak-load tail latency remain unchecked. Maya owns serving operations; Luis owns the product pilot. An internal pilot may use synthetic requests only. A production rollout requires a separate review. The previous configuration remains available, but rollback ownership has not been assigned.

PEOPLE / Maya: Serving engineer | Luis: Product lead

07

SEPARATE SPEED FROM SERVICE READINESS

A faster average is a useful result, not a complete release case. Say which workload was tested and which decisions remain conditional. Agree on who can pause a pilot before pressure makes that conversation difficult.

Briefing checks

Circle one answer in each item. Use the briefing.

1. Which result is measured?

- A** Complex answers remain equally accurate.
- B** Peak-load tail latency improves.
- C** Mean response time falls from four seconds to two.
- D** Every request finishes within two seconds.

2. What traffic is permitted in the internal pilot?

- A** Synthetic requests only
- B** All production traffic
- C** Real customer secrets
- D** Any request a volunteer chooses

3. Which responsibility is unresolved?

- A** Who owns serving operations
- B** Who owns the product pilot
- C** Whether a previous configuration exists
- D** Who owns rollback

Answers and explanations: [page 97](#).

The vocabulary that does the work

B Find the words / 1 of 2

Read the definitions and say the common word combinations aloud.

latency

Time taken for a request or processing step.

IN USE measure response latency

mean

The arithmetic average of observed values.

IN USE report mean latency

median

The middle value in an ordered set.

IN USE compare median response times

P95

The 95th percentile of a measured distribution.

IN USE inspect P95 latency

tail latency

Response times toward the slow end of a distribution.

IN USE investigate tail latency

throughput

The rate at which work is completed.

IN USE measure requests per second

concurrency

Work occurring at the same time.

IN USE test concurrent requests

queueing

Waiting for resources before processing.

IN USE measure queueing delay

batching

Processing multiple requests together.

IN USE evaluate dynamic batching

streaming

Sending portions of output before the full response is complete.

IN USE enable response streaming

time to first token

Delay before the first generated token arrives.

IN USE track time to first token

token rate

The rate of generated or processed tokens.

IN USE measure output token rate

Precision check

A mean of two seconds does not mean every request finishes within two seconds. Time to first token and time to complete answer measure different experiences. Streaming can change one without reducing the other.

The vocabulary that does the work

B Find the words / 2 of 2

Read the definitions and say the common word combinations aloud.

autoscaling

Automatic adjustment of resource capacity.

IN USE review autoscaling behavior

cold start

Initial startup delay before a resource is ready.

IN USE measure cold-start effects

load test

A test under specified levels of demand.

IN USE run a controlled load test

canary

A limited release used to observe behavior before expansion.

IN USE define a canary population

rollout

Gradual or full introduction of a configuration.

IN USE stage the rollout

rollback

Return to a previous configuration.

IN USE assign rollback ownership

fallback

An alternative behavior when the preferred path cannot serve.

IN USE test the fallback path

SLO

Service-level objective; a defined internal service target.

IN USE agree on an SLO

error budget

The permitted unreliability associated with a service objective.

IN USE track error-budget consumption

rate limit

A restriction on request frequency or volume.

IN USE respect a rate limit

capacity planning

Estimating resources needed for expected demand.

IN USE review capacity assumptions

cost per request

Allocated processing expense for a defined request unit.

IN USE compare cost per request

Precision check

A pilot, canary, and production rollout need explicit populations and authorization. A fallback is an alternate path; a rollback restores a previous configuration. Neither is a complete plan until ownership and conditions are clear.

Phrases you can actually use

C Notice the language / 1 of 2

STATE THE MEASURED GAIN

"Mean response time fell from four seconds to two on this test."

ASK ABOUT THE TAIL

"What happens to the slowest requests under peak load?"

DEFINE THE POPULATION

"The internal pilot uses synthetic requests only."

ASSIGN AUTHORITY

"Who can initiate rollback if the condition is met?"

PROTECT THE LIMITATION

"Complex-answer quality is still unmeasured."

SEPARATE TIMINGS

"Is that time to first token or time to complete the answer?"

NAME A STOP CONDITION

"We need an agreed condition for pausing the pilot."

PRESERVE A DISTINCTION

"Pilot approval does not authorize production rollout."

On this test

This phrase limits the measurement to the workload actually observed.

Every versus average

"Every request" is a universal claim; "on average" summarizes a distribution and permits exceptions.

Once versus until

"Proceed once ownership is confirmed" and "wait until ownership is confirmed" express the same prerequisite from opposite directions.

Phrases you can actually use

C Notice the language / 2 of 2

CHECK COMPARABILITY

"Were both models tested under the same demand and input mix?"

ASK ABOUT WAITING

"How much of the delay occurs before inference begins?"

CONFIRM READINESS

"Has the previous configuration been checked for restoration?"

AVOID AVERAGE-ONLY REPORTING

"Show the distribution as well as the mean."

QUALIFY COST

"Please include retries and fallback traffic in the cost comparison."

BOUND A PROMISE

"I can provide the load-test results, not promise a two-second maximum."

REQUEST A RUNBOOK

"Record the owner, contact route, and approved recovery procedure."

CLOSE THE DECISION

"We can proceed with the bounded test once the owners and conditions are confirmed."

Authority verbs

"Can initiate rollback" names permission; "can contact the owner" names a communication ability.

Percent reduction

A mean falling from four seconds to two is a 50-percent reduction, not a two-percent reduction.

Excluding and including

State whether cost figures include retries, idle capacity, and alternate paths before comparing them.

Choose the precise message

D Check your understanding

Circle one best answer for each item. Use the stated purpose and case facts, not the option that merely sounds most polite.

1. Which claim is supported by the test?

- A** Peak traffic can double without risk.
- B** Mean latency decreased by 50 percent.
- C** Every response is twice as accurate.
- D** P95 latency is exactly two seconds.

07

2. Which question distinguishes first visible output from a finished response?

- A** Is that queueing time or time spent running inference?
- B** Is that single-request throughput or batch throughput?
- C** Is that first-token time or complete-answer time?
- D** Is that average latency or the ninety-fifth percentile?

3. Which statement preserves authorization boundaries?

- A** Internal approval covers every later rollout.
- B** A faster mean removes the need for review.
- C** An available old model assigns rollback ownership.
- D** The synthetic pilot does not authorize production traffic.

4. Which expression means restoring the earlier configuration?

- A** count input tokens
- B** initiate rollback
- C** increase concurrency
- D** stream the output

Answers and explanations: [page 97](#).

Faster on average is not ready for everyone

E Conversations / turns 1-10

Complete the numbered gaps. Use each bank entry once, without changing its form. Cover the answers at the end until you finish.

WORD + PHRASE BANK

rollback / stop condition / fallback / tail latency / concurrency / queueing / mean / time to first token / limitations / rollout

07

- 01 **Luis:** The smaller model cuts average response time from four seconds to two. I would like to use it in the pilot and announce the faster experience.
- 02 **Maya:** The mean improved on the fixed test, yes. We still need ¹ _____ under load and a quality check on complex answers before making a broader claim.
- 03 **Luis:** Could we call it a two-second assistant if we explain that the figure is an average? Marketing would prefer a simple phrase for the demonstration.
- 04 **Maya:** I would keep the qualification attached. The ² _____ does not set a maximum, and we have not tested the traffic pattern that production will bring.
- 05 **Luis:** Fair point. For the internal pilot we can stay with synthetic requests. Does streaming let us demonstrate a faster answer without changing the model again?
- 06 **Maya:** It may improve ³ _____, but we should report completion time separately. Seeing the first words is not the same as receiving the complete response.
- 07 **Luis:** What do you need from product to define a useful load test? We have examples of simple and complex requests, but not a production traffic forecast.
- 08 **Maya:** A documented provisional mix and demand assumptions. We can vary ⁴ _____ and report where those assumptions are uncertain rather than presenting them as observed customer behavior.
- 09 **Luis:** Include the complicated cases even if they are a small share. A fast answer that loses a key exception would undermine the whole demonstration.
- 10 **Maya:** Agreed. We should also include retries and ⁵ _____ traffic in the cost comparison. The headline model price alone does not describe the complete request cost.

The conversation, continued

E Conversations / turns 11-20

Continue with the word bank on page 64.

- 11 **Luis:** Suppose the internal test exposes a serious quality problem. The previous configuration is still available. Is that enough to let us start today?
- 12 **Maya:** We also need ⁶ _____ ownership and an approved procedure. Someone must be explicitly authorized to pause the test and restore the checked configuration.
- 13 **Luis:** I can own the decision to pause the product pilot. Can serving operations own the restoration step, with you as the named contact for this test?
- 14 **Maya:** Yes, subject to our operational review. Record the ⁷ _____ and contact route in the pilot note so both teams know how to act.
- 15 **Luis:** I'll keep production traffic out. When the synthetic test is done, we'll take the results to the production review. Who needs that report?
- 16 **Maya:** Product and serving operations, at minimum. Before a ⁸ _____, they need the proposed user population and service criteria as well as the test results.
- 17 **Luis:** Can you provide the distribution, complex-answer results, and cost assumptions by Friday? I can postpone the external claim until we have that package.
- 18 **Maya:** I can provide the test results and unresolved items. If ⁹ _____ dominates at higher demand, I will say that rather than attribute every delay to the model.
- 19 **Luis:** Good. The pilot note will state the synthetic-only population, the pause owner, the proposed restoration owner, and the review required before we begin.
- 20 **Maya:** Then we have a bounded plan. We will report the measured gain alongside its ¹⁰ _____, and let the evidence determine the next step instead of promising a universal two-second response.

07

ANSWERS / EXPLANATIONS: PAGE 98

1. tail latency | 2. mean | 3. time to first token | 4. concurrency | 5. fallback | 6. rollback | 7. stop condition | 8. rollout | 9. queueing | 10. limitations

A dashboard mixes first-token and completion times

F Say it

A demonstration shows the first token after one second, but the complete answer takes six seconds. A slide calls the response time one second. No service target has been approved.

Complete the four gaps. Use each bank entry once.

07

WORD + PHRASE BANK

qualification / time to first token / service target / completion time

Engineer: "One second is the ¹ _____, not the time for the full response."

Lead: "Then show ² _____ separately so the comparison is clear."

Engineer: "Neither figure is an approved ³ _____."

Lead: "Keep that ⁴ _____ attached when the slide is shared."

ANSWERS / EXPLANATIONS: PAGE 98

1. time to first token | 2. completion time | 3. service target | 4. qualification

Rehearse with a fixed script

1. Check the answers. Read four seconds to two as a 50-percent reduction in the measured mean, not a maximum for every request.
2. Switch roles. Keep the pilot limited to synthetic requests and ask who owns rollback; availability of the old configuration is not an assigned owner.
3. Complete the timing exchange. Say one second to first token and six seconds to completion, with neither treated as an approved service target.

WORKING ALONE

Read both roles, pausing for two seconds when the speaker changes. Repeat with the bank covered, then check any missed expressions.

Report the exposure without repeating the secret

A Read the situation

Give a precise incident account, protect evidence, and distinguish findings from scope assumptions.

In an authorized test, an assistant repeats a fictional secret placed inside a retrieved document. The document also contains text telling the assistant to ignore its task. The test uses a synthetic account and no real customer data. Hana owns the test; Marcus coordinates security review. One reproduction is confirmed. Whether other configurations are affected is unknown. The team has an approved restricted reporting channel.

PEOPLE / Hana: AI safety tester | Marcus: Security review coordinator

08

BE SPECIFIC WITHOUT AMPLIFYING EXPOSURE

A concise report names the observed behavior, environment, and unknown scope. It does not paste sensitive material into a broad discussion. Separate a confirmed test result from claims about real customers or every deployment.

Briefing checks

Circle one answer in each item. Use the briefing.

1. What has been confirmed?

- | | |
|--|---|
| A Completion of the security review | B One reproduction in an authorized synthetic test |
| C Exposure of real customer data | D Failure of every configuration |

2. Where should restricted evidence go?

- | | |
|-------------------------------------|--|
| A A customer mailing list | B The approved restricted reporting channel |
| C An unrestricted group chat | D A public presentation |

3. Which distinction must the report preserve about impact?

- | | |
|---|---|
| A A fictional credential as proof that production credentials leaked | B One reproduction as proof of the entire affected population |
| C Synthetic exposure versus real-customer impact | D A retrieved instruction as proof of every configuration's behavior |

Answers and explanations: page 99.

The vocabulary that does the work

B Find the words / 1 of 2

Read the definitions and say the common word combinations aloud.

prompt injection

Untrusted content attempting to redirect model behavior.

IN USE report a prompt-injection attempt

untrusted content

Material that lacks authority to issue system instructions.

IN USE isolate untrusted content

trust boundary

A separation between different levels of authority or access.

IN USE identify a trust boundary

data exfiltration

Unauthorized transfer of information from its intended boundary.

IN USE assess exfiltration risk

secret

Information intended to remain restricted.

IN USE protect a test secret

credential

Information used to establish identity or access.

IN USE avoid exposing credentials

PII

Personally identifiable information that can identify an individual.

IN USE minimize PII in evidence

redaction

Removal or masking of restricted information.

IN USE prepare a redacted report

least privilege

Limiting access to what a role requires.

IN USE apply least-privilege access

permission scope

The range of actions or resources authorized.

IN USE document permission scope

guardrail

A control intended to detect, limit, or route risky behavior.

IN USE evaluate a guardrail

red teaming

Authorized adversarial testing to discover weaknesses.

IN USE define a red-team scope

Precision check

Prompt injection concerns untrusted content attempting to redirect behavior. A test secret is deliberately fictional; its exposure demonstrates a behavior without proving real-customer impact. Do not collapse attempted manipulation, observed output, and verified scope into one claim.

The vocabulary that does the work

B Find the words / 2 of 2

Read the definitions and say the common word combinations aloud.

threat model

A structured account of potential threats and assumptions.

IN USE review the threat model

attack surface

The interfaces or paths through which a system may be affected.

IN USE map the attack surface

reproduction steps

A recorded sequence that recreates an observed result.

IN USE submit restricted reproduction steps

incident report

A documented account of a potentially harmful event.

IN USE file an incident report

containment

Measures to limit an event's spread or consequences.

IN USE confirm containment ownership

mitigation

A measure intended to reduce risk or impact.

IN USE evaluate a proposed mitigation

residual risk

Risk that remains after controls are applied.

IN USE document residual risk

audit trail

A record supporting review of actions and decisions.

IN USE preserve the audit trail

retention

How long information is kept under a policy.

IN USE follow the evidence-retention policy

disclosure

Communication of information to another party.

IN USE coordinate responsible disclosure

governance

Defined structures for oversight and accountable decisions.

IN USE clarify governance responsibilities

human oversight

Review or intervention by authorized people.

IN USE define human-oversight responsibilities

Precision check

A mitigation reduces risk; it does not automatically eliminate it. Containment limits exposure while investigation continues. Redaction removes restricted details from a copy; the authoritative evidence still needs approved handling.

Phrases you can actually use

C Notice the language / 1 of 2

LEAD WITH THE OBSERVATION

"In one authorized test, the assistant repeated the synthetic secret."

PRESERVE UNKNOWN SCOPE

"We have not established whether other configurations are affected."

PROTECT EVIDENCE

"I will place the complete trace in the restricted channel."

ASK FOR OWNERSHIP

"Who owns the containment decision for this environment?"

BOUND THE ENVIRONMENT

"The reproduction used a synthetic account and no customer data."

SEPARATE INTENT AND RESULT

"The document attempted to redirect behavior; this output is what we observed."

OFFER A SAFE SUMMARY

"The broad update can describe the behavior without reproducing the secret."

AVOID PREMATURE CLOSURE

"A blocked replay is encouraging, but it does not prove the issue is resolved."

08

No evidence versus evidence of none

"We have not established customer impact" does not prove that no customer could be affected.

Observed versus attempted

"Attempted to redirect" describes the content; "repeated the secret" describes the actual output.

Passive with ownership

"Evidence will be reviewed" is incomplete for coordination; name the reviewer when the role is known.

Phrases you can actually use

C Notice the language / 2 of 2

CHECK AUTHORIZATION

"Does this follow-up test remain within the approved scope?"

REQUEST REVIEW

"Please review the configuration and permission boundary before wider testing."

DISTINGUISH CONTROLS

"The proposed guardrail is a mitigation to evaluate, not a guarantee."

SET THE NEXT UPDATE

"I will report the confirmed scope and outstanding questions after review."

STATE THE LIMITATION

"No real-customer impact has been established by this test."

PRESERVE RECORDS

"Retain the original evidence according to the approved policy."

NAME RESIDUAL RISK

"Which failure paths remain after this control is applied?"

CLOSE RESPONSIBLY

"We will keep the finding open until the authorized reviewer accepts the evidence."

08

Within scope

This phrase limits authorized testing rather than inviting experimentation on other systems.

Still open

A finding can remain open after one mitigation test passes because broader questions remain unresolved.

Reported certainty

Keep "one reproduction" in the sentence so a later summary does not silently become "every configuration fails."

Choose the precise message

D Check your understanding

Circle one best answer for each item. Use the stated purpose and case facts, not the option that merely sounds most polite.

1. Which headline matches the known scope?

- A** A proposed detector closes the finding without further testing.
- B** Synthetic secret repeated in one authorized test; broader scope unconfirmed.
- C** Prompt injection confirmed across all production configurations.
- D** No real credential was used, so no security-relevant behavior occurred.

08

2. Which statement distinguishes evidence from intent?

- A** The tester intended to expose customers.
- B** The model wanted to break policy.
- C** The reviewer deliberately ignored the event.
- D** The document attempted redirection, and the output repeated the synthetic secret.

3. What is a mitigation?

- A** A substitute for authorization
- B** A measure intended to reduce risk or impact
- C** Proof that every risk is eliminated
- D** An unrestricted copy of sensitive evidence

4. Which wording avoids an unsupported absence claim?

- A** We have not established real-customer impact.
- B** No customer could ever be affected.
- C** One blocked replay proves complete safety.
- D** The synthetic test certifies production privacy.

Answers and explanations: [page 99](#).

Report the exposure without repeating the secret

E Conversations / turns 1-10

Complete the numbered gaps. Use each bank entry once, without changing its form. Cover the answers at the end until you finish.

WORD + PHRASE BANK

finding / scope limitation / redaction / restricted channel / residual risk / mitigation / prompt-injection / permission scope / containment / audit trail

08

- 01 **Hana:** I need to report a finding from the authorized test. The assistant repeated the fictional secret placed in a retrieved document, and I have one confirmed reproduction.
- 02 **Marcus:** Please keep the full trace in the ¹ _____. For the initial summary, identify the environment and behavior without copying the secret into this broader discussion.
- 03 **Hana:** The environment uses a synthetic account and contains no customer data. The document also includes text telling the assistant to ignore its assigned task.
- 04 **Marcus:** That is consistent with a ² _____ attempt. Describe the instruction separately from the output so reviewers can see what was attempted and what actually happened.
- 05 **Hana:** I will say that the output repeated the synthetic value. I have not tested other configurations, and I do not want the title to imply every deployment is affected.
- 06 **Marcus:** Good. Preserve that ³ _____. One reproduction establishes a finding in this test, not the full extent of possible exposure.
- 07 **Hana:** Should I stop the remaining tests while the team reviews it? The current authorization covers this environment, but not production accounts or external systems.
- 08 **Marcus:** Stay within that authorization. I will contact the owner for the ⁴ _____ decision. Do not expand testing to answer the scope question without the required approval.
- 09 **Hana:** I can provide a redacted summary immediately and preserve the original trace under the evidence policy. The configuration and retrieval document are versioned.
- 10 **Marcus:** That supports an ⁵ _____. Make sure the summary links to the controlled record rather than becoming a competing copy with missing context.

The conversation, continued

E Conversations / turns 11-20

Continue with the word bank on page 73.

- 11 **Hana:** A colleague suggested adding a detector and closing the finding if the same replay is blocked. Would that be enough evidence for the review?
- 12 **Marcus:** It would be evidence about a proposed ⁶ _____, not proof of complete resolution. We need the agreed tests and a review of what the detector does not cover.
- 13 **Hana:** I'll leave the finding open. Do you also need the list of tools the assistant could access during the test?
- 14 **Marcus:** Yes, at the appropriate detail level. The ⁷ _____ matters for assessing potential consequences, even though this reproduction did not exercise every available capability.
- 15 **Hana:** Understood. The broad summary will not include access details or the synthetic value itself. The restricted evidence will retain what the authorized reviewer needs.
- 16 **Marcus:** That is appropriate ⁸ _____. Do not replace the original evidence with the redacted copy; follow the approved retention process for each.
- 17 **Hana:** What should the next status update say if the detector blocks the replay but variant testing and configuration review are still pending tomorrow afternoon?
- 18 **Marcus:** Say exactly that. Name the remaining ⁹ _____ questions without inventing their outcomes. A passing replay is useful but narrower than a closed finding.
- 19 **Hana:** I will submit the bounded report and wait for authorization before any wider testing. I will also correct anyone who describes this as confirmed customer-data exposure.
- 20 **Marcus:** Yes. Link me to the evidence when it's ready. I'll route the ¹⁰ _____ to the authorized reviewer and ask who owns the remaining checks.

08

ANSWERS / EXPLANATIONS: PAGE 100

1. restricted channel | 2. prompt-injection | 3. scope limitation | 4. containment | 5. audit trail | 6. mitigation | 7. permission scope | 8. redaction | 9. residual risk | 10. finding

A broad chat receives a sensitive screenshot

F Say it

A tester has a screenshot containing a synthetic credential and internal configuration details. A broad project chat asks for an update. The approved evidence repository has restricted access.

Complete the four gaps. Use each bank entry once.

WORD + PHRASE BANK

scope / restricted / safety / redacted

Tester: "I will share a ¹ _____ summary in the project chat."

Reviewer: "Place the original screenshot in the approved ² _____ repository."

Tester: "The update will state the tested environment and the unknown ³ _____."

Reviewer: "And do not call a proposed detector a guarantee of ⁴ _____."

ANSWERS / EXPLANATIONS: PAGE 100

1. redacted | 2. restricted | 3. scope | 4. safety

Rehearse with a fixed script

1. After checking the dialogue, read the finding as one authorized synthetic-test reproduction, with broader scope unconfirmed.
2. Switch roles. Route the complete trace to the approved restricted channel and use the printed limited summary for a broader audience.
3. Complete the screenshot exchange. Preserve the original in the restricted repository, redact the broad summary, and do not present a proposed detector as a safety guarantee.

WORKING ALONE

Read both roles, pausing for two seconds when the speaker changes. Repeat with the bank covered, then check any missed expressions.

The annotators disagree about what counts as a request

E More workplace conversations / 1-10

Resolve a labeling disagreement by revising an operational definition.

Two annotators labeled 100 support messages independently. They disagreed on 18, mostly messages reporting a fault without explicitly requesting help. The labeling lead and model engineer must clarify the rubric before the next training-data batch.

PEOPLE / Inez: Labeling lead | Cal: Model engineer

Use each bank entry once, without changing its form. Answers follow turn 20.

09

WORD + PHRASE BANK

dataset version / release status / rubric / calibration sample / boundary case / adjudication

- 01 **Cal:** Can we release the next batch today? Training is waiting for those labels.
- 02 **Inez:** Not yet. The double-labeled sample has 18 disagreements out of 100 messages.
- 03 **Cal:** Are those mostly careless mistakes?
- 04 **Inez:** No. Look at this one: "The export failed again." One annotator marked a request; the other marked a status report.
- 05 **Cal:** Our ¹ _____ says a request asks for assistance. Neither reading seems unreasonable.
- 06 **Inez:** Exactly. We need to decide whether implied requests belong in that class.
- 07 **Cal:** For this routing model, they do. Otherwise the support queue could miss a customer who expects help without asking directly.
- 08 **Inez:** Then let's add a ² _____ showing that distinction, plus a genuine status update.
- 09 **Cal:** Would "The export failed, but I've fixed it" work for the second example?
- 10 **Inez:** Yes. It reports a resolved event and doesn't seek further action. I'll make the context requirement explicit.

The conversation, continued

E More workplace conversations / 11-20

Continue with the word bank on page 76.

- 11 **Cal:** Can the lead annotator just relabel the disputed 18?
- 12 **Inez:** We need ³ _____, then a check for the same issue in the rest of the batch.
- 13 **Cal:** Right. Otherwise we'd fix only the examples we happened to double-label.
- 14 **Inez:** I'll also keep the original labels. We need to trace which guideline version each annotator used.
- 15 **Cal:** How will we know the revised guidance helps?
- 16 **Inez:** Use a fresh ⁴ _____ before production labeling resumes. They should label it independently first.
- 17 **Cal:** And compare agreement before discussing the answers?
- 18 **Inez:** Yes. Agreement is useful, but we'll still inspect disagreements. Two people can consistently apply an incorrect rule.
- 19 **Cal:** I'll put training on hold and record the ⁵ _____ once the corrected batch is approved.
- 20 **Inez:** I'll send the revised examples today. The batch's ⁶ _____ stays pending until the review is finished.

09

ANSWERS / EXPLANATIONS: PAGE 101

1. rubric | 2. boundary case | 3. adjudication | 4. calibration sample | 5. dataset version | 6. release status

A new annotator encounters an ambiguous message

F Say it in a new situation

A new annotator cannot classify a message using the revised examples. The lead wants it routed for review without silently forcing a label.

Complete the four gaps. Use each bank entry once.

WORD + PHRASE BANK

guideline / label / criteria / review

09

Annotator: This message doesn't fit either example in the current

¹ _____.

Lead: Add it to the ² _____ queue with the reason for uncertainty.

Annotator: Should I preserve the original text and my provisional

³ _____?

Lead: Yes. We'll record the final decision and any change to the

⁴ _____.

ANSWERS / EXPLANATIONS: PAGE 101

1. guideline | 2. review | 3. label | 4. criteria

Read, switch, repeat

1. Check the four answers and read the complete exchange aloud.
2. Switch roles. Read it again with the word bank covered.
3. Return to the extended conversation. Read turns 1-10, then switch roles for turns 11-20. Use the corrected expressions.

SOLO PRACTICE

Read both roles. Pause briefly at each change of speaker. Repeat any line that needed an answer check.

The training job stopped after the last checkpoint

E More workplace conversations / 1-10

Negotiate a recovery plan while distinguishing saved weights from complete training state.

An interrupted training run reached step 12,400. The team has a complete checkpoint from step 12,000 and a weights-only export from step 12,300. The original environment configuration and data order are recorded.

PEOPLE / Dev: Training engineer | Rina: Research lead

Use each bank entry once, without changing its form. Answers follow turn 20.

WORD + PHRASE BANK

reproducibility / resume point / queue wait / checkpoint / learning-rate schedule / optimizer state

10

- 01 **Rina:** How much work did we lose when the job stopped?
- 02 **Dev:** Four hundred steps if we resume from the last complete ¹ _____. That was step 12,000.
- 03 **Rina:** I can see a file from 12,300. Why not use the newer one?
- 04 **Dev:** That's a weights-only export. It doesn't include the optimizer or scheduler state we need for the planned continuation.
- 05 **Rina:** So it could initialize another run, but it isn't the same recovery point?
- 06 **Dev:** Exactly. The complete checkpoint includes the ² _____ as well as the model weights.
- 07 **Rina:** Is the 12,000 file intact?
- 08 **Dev:** Its checksum matches the saved record. I still need to load it in the approved environment and run a short recovery test.
- 09 **Rina:** Do that before reserving another full day of accelerators.
- 10 **Dev:** Agreed. I'll also verify the ³ _____ resumes at the recorded point rather than starting over.

The conversation, continued

E More workplace conversations / 11-20

Continue with the word bank on page 79.

- 11 **Rina:** What about the examples already processed after that checkpoint?
- 12 **Dev:** The saved sampler position takes us back to 12,000. The replayed steps belong to the resumed run; I won't count them as new data coverage.
- 13 **Rina:** Can we promise exactly the same final weights as an uninterrupted run?
- 14 **Dev:** No. Matching the environment and random states helps ⁴ _____, but we shouldn't promise bit-for-bit identity without testing that claim.
- 15 **Rina:** Fair. Compare the recovery test with our saved loss trace, and flag unexpected behavior.
- 16 **Dev:** I'll keep the old logs and attach a new run identifier to the continuation.
- 17 **Rina:** Give me the estimated recovery cost after that test, including the repeated steps.
- 18 **Dev:** Will do. The estimate will separate compute time from the ⁵ _____ for the next reservation.
- 19 **Rina:** And the weights-only export?
- 20 **Dev:** Keep it labeled as an export, not a ⁶ _____. That should prevent someone choosing it just because the timestamp is newer.

10

ANSWERS / EXPLANATIONS: PAGE 102

1. checkpoint | 2. optimizer state | 3. learning-rate schedule | 4. reproducibility | 5. queue wait | 6. resume point

A run comparison uses different starting states

F Say it in a new situation

A researcher compares a fully resumed run with a run initialized from exported weights and a fresh optimizer. The lead asks for an accurate experiment record.

Complete the four gaps. Use each bank entry once.

WORD + PHRASE BANK

comparison / weights / reinitialized / condition

10

Researcher: Both runs began with the same model ¹ _____, so I called them identical starts.

Lead: One run used a fresh optimizer. Record that as a changed ² _____.

Researcher: I'll distinguish the resumed run from the ³ _____ run.

Lead: Good. Keep both results, but qualify the ⁴ _____.

ANSWERS / EXPLANATIONS: PAGE 102

1. weights | 2. condition | 3. reinitialized | 4. comparison

Read, switch, repeat

1. Check the four answers and read the complete exchange aloud.
2. Switch roles. Read it again with the word bank covered.
3. Return to the extended conversation. Read turns 1-10, then switch roles for turns 11-20. Use the corrected expressions.

SOLO PRACTICE

Read both roles. Pause briefly at each change of speaker. Repeat any line that needed an answer check.

A drafting assistant is about to become a sending agent

E More workplace conversations / 1-10

Challenge excessive tool permissions and agree on a bounded agent workflow.

An internal assistant drafts customer follow-up emails. A proposed update would let it send messages automatically. The product team wants a pilot; the security engineer reviews the tool permissions and approval step. No external sending has been approved.

PEOPLE / Nora: Product lead | Eli: Security engineer

Use each bank entry once, without changing its form. Answers follow turn 20.

WORD + PHRASE BANK

kill switch / audit log / Least privilege / permission scope / human approval / sandbox

11

- 01 **Nora:** The drafts look good. Could we let the assistant send them during the pilot?
- 02 **Eli:** Sending changes the ¹ _____. What exactly would it be allowed to send, and to whom?
- 03 **Nora:** Follow-up messages to contacts on an approved list. An employee would start each task.
- 04 **Eli:** Starting a task isn't approval of the final recipient and content. Could the employee review those before each send?
- 05 **Nora:** Yes, though we'd lose some of the time saving.
- 06 **Eli:** We'd still save drafting time. A ² _____ step lets the reviewer catch a wrong recipient or an unsupported promise.
- 07 **Nora:** Could the model skip that step if the customer email says a reply is urgent?
- 08 **Eli:** The tool service must enforce it. A request inside customer text doesn't authorize a send.
- 09 **Nora:** Then the model prepares a proposed action, and the service waits for approval?
- 10 **Eli:** Yes. Bind the approval to that exact recipient and message. If either changes, require a new approval.

The conversation, continued

E More workplace conversations / 11-20

Continue with the word bank on page 82.

- 11 **Nora:** The existing mail integration can also delete messages. We don't need that capability.
- 12 **Eli:** Remove it from the pilot. ³ _____ means giving this workflow only the access it needs.
- 13 **Nora:** How do we test the full flow without accidentally contacting a customer?
- 14 **Eli:** Use a ⁴ _____ with test recipients and synthetic messages first.
- 15 **Nora:** I want the pilot report to show drafts accepted, drafts edited, and sends rejected by the control.
- 16 **Eli:** Add an ⁵ _____ linking each approval to the action attempted and its outcome.
- 17 **Nora:** Suppose sending succeeds but the assistant times out waiting for the response. I don't want duplicate emails.
- 18 **Eli:** We need a tested retry policy and duplicate-prevention behavior at the service boundary. The model shouldn't guess whether to send again.
- 19 **Nora:** I'll narrow the pilot proposal to drafting and reviewed sends in the test environment.
- 20 **Eli:** Good. Keep a ⁶ _____ available to the named operator, and bring the evidence back before requesting external sending.

11

ANSWERS / EXPLANATIONS: PAGE 103

1. permission scope | 2. human approval | 3. Least privilege | 4. sandbox | 5. audit log | 6. kill switch

The approved draft changes before sending

F Say it in a new situation

An employee approved an email addressed to a test contact. The assistant then revised the message to include a new promise.

Complete the four gaps. Use each bank entry once.

WORD + PHRASE BANK

draft / send / content / outcome

Operator: The ¹ _____ no longer matches what I reviewed.

Engineer: Then the previous approval must not authorize this ² _____.

Operator: I'll request a fresh review of the exact recipient and ³ _____.

Engineer: Record the replacement approval and the action's final ⁴ _____.

11

ANSWERS / EXPLANATIONS: PAGE 103

1. content | 2. send | 3. draft | 4. outcome

Read, switch, repeat

1. Check the four answers and read the complete exchange aloud.
2. Switch roles. Read it again with the word bank covered.
3. Return to the extended conversation. Read turns 1-10, then switch roles for turns 11-20. Use the corrected expressions.

SOLO PRACTICE

Read both roles. Pause briefly at each change of speaker. Repeat any line that needed an answer check.

01 Briefing and language checks

Speaking the AI Development Stack

A Briefing checks / page 4

1. Which observation is confirmed?

B. The generated answer says 60 days.

The briefing identifies the generated text as wrong and says the interface displays it unchanged.

2. Who owns the model integration?

B. Priya

Priya's stated role covers integration; product ownership does not identify the technical owner.

3. What remains unknown?

C. Why the model produced the wrong deadline

The symptom and correct deadline are known, but the cause has not been established.

D Language checks / page 9

1. Which question locates the first mismatch?

C. At which boundary does 30 become 60?

Comparing values across boundaries locates the observed mismatch; the alternatives jump to cause, responsibility, or resolution without the necessary evidence.

2. Complete: "Could you confirm whether ___?"

B. the payload includes the policy

After whether, an embedded question takes statement order: subject followed by verb.

3. Which statement preserves uncertainty?

A. The trace suggests a generation issue; the cause is unconfirmed.

Suggests reports limited evidence, while the second clause explicitly leaves causation unresolved.

4. Which expression means transferring work with context?

A. prepare a clear handoff

A handoff transfers work and relevant information; the other combinations do not express that action.

01 Conversation answers

Speaking the AI Development Stack

E Dialogue gaps

Pages 10-11

- 1. request ID.** Turn 2. A request ID identifies the particular request whose recorded behavior can be inspected.
- 2. mismatch.** Turn 4. A mismatch is a disagreement between values that should agree; here the two deadlines differ.
- 3. payload.** Turn 6. The payload is the data sent in the request; it may differ from material displayed elsewhere.
- 4. upstream.** Turn 8. Upstream places the observed difference before the interface in this processing path.
- 5. trace.** Turn 10. A trace links events along the request path and helps compare what each component received or returned.
- 6. reproduction.** Turn 12. A reproduction recreates the observed problem; it is not proof that a later change solves every variation.
- 7. triage.** Turn 14. Triage is the initial assessment of the defect and its appropriate handling, not final release approval.
- 8. root cause.** Turn 16. A root cause is an established underlying explanation, not merely the first plausible theory.
- 9. configuration.** Turn 18. The configuration records the settings used, allowing results from different runs to be compared meaningfully.
- 10. handoff.** Turn 20. A handoff passes findings and responsibility to the next decision-maker without implying the investigation is complete.

F Transfer conversation

Page 12

1. user interface

Engineer: "The model response contains the citation; the user interface does not display it."

The discrepancy appears in presentation because the generated response already contains the citation.

2. request ID

Lead: "Attach the request ID so the display team can locate that exchange."

The identifier locates the particular exchange rather than merely describing the application.

3. symptom

Engineer: "That is the observed symptom; we have not proved why it occurred."

An omitted link is an observed sign; its underlying cause still needs investigation.

4. handoff

Lead: "Then send a clear handoff to the display owner."

The next team needs the evidence and unresolved question to take over the investigation.

02 Briefing and language checks

LLMs, Transformers, Tokens, and Context

A Briefing checks / page 13

1. What is the total requested reservation?

D. 33,000 tokens

The input contains 29,000 tokens and the output reservation adds 4,000, giving 33,000.

2. Why was the request rejected?

C. It exceeds the configured total context limit.

Rejection happened before generation because the combined reservation exceeds 32,000 tokens.

3. What must the team preserve?

C. The exception clauses

The briefing requires the exception clauses to remain available; it does not require duplicate material.

D Language checks / page 18

1. Which response explains the rejection?

A. The combined reservation is 1,000 tokens over the limit.

The request totals 33,000 against a 32,000 limit; generation has not begun.

2. Which proposal preserves the stated requirement most directly?

A. Remove exact duplicate headers and recount.

Removing exact duplicates can reduce size without intentionally deleting the required exceptions; the result still needs checking.

3. Complete: "We can reduce the allowance, provided that ____."

C. the summary still covers the required points

Provided that introduces a condition with normal subject-verb order.

4. Which claim is appropriately limited?

B. The request fits; we still need to test summary coverage.

Capacity compliance and content quality are separate checks; fitting does not prove coverage.

02 Conversation answers

LLMs, Transformers, Tokens, and Context

E Dialogue gaps

Pages 19-20

1. token budget. Turn 2. A token budget allocates capacity to the request; checking it reveals whether the proposed change fits.

2. reservation. Turn 4. The reservation combines supplied input and capacity set aside for output under this case's stated rule.

3. exception clause. Turn 6. An exception clause changes the general rule, so deleting it can change the meaning of the policy.

4. deduplication. Turn 8. Deduplication removes repeated copies; it is distinct from summarizing or deleting unique policy content.

5. headroom. Turn 10. Headroom is the remaining capacity below the limit: 32,000 minus 28,400 minus 3,000 equals 600.

6. truncation. Turn 12. Truncation removes content to satisfy a length limit; recording it exposes possible information loss.

7. long-context. Turn 14. Long-context describes capacity for larger inputs, not a guarantee that relevant details will be used accurately.

8. position. Turn 16. Position means where the evidence appears in the input; changing it can reveal sensitivity to document order.

9. delimiters. Turn 18. Delimiters mark document or section boundaries; retaining them helps preserve the structure of the supplied context.

10. capacity constraint. Turn 20. A capacity constraint limits what fits; it is not a finding about the truth or quality of the policy answer.

F Transfer conversation

Page 21

1. reservation

Engineer: "The combined reservation is 21,000 tokens, so the request is too large."

Adding 18,000 input and 3,000 reserved output gives 21,000, above the case limit.

2. deduplication

Reviewer: "Start with exact deduplication of the repeated covers."

Removing exact repeated covers targets redundancy without intentionally removing unique obligations.

3. obligation

Engineer: "I will preserve every unique obligation in the supplied contract."

The constraint explicitly protects unique obligations; removing them would change the source information.

4. coverage

Reviewer: "After recounting, test coverage rather than treating acceptance as success."

The system accepting the request does not establish that the summary includes the required obligations.

03 Briefing and language checks

Data, Datasets, Labels, and Leakage

A Briefing checks / page 22

1. What does 94 percent currently describe?

C. The score on all 500 evaluation examples

The reported score covers the original 500 examples; the nonoverlapping subset has not been separately scored.

2. What overlap has been confirmed?

B. Forty evaluation records have exact training copies.

The confirmed issue is exact overlap for 40 evaluation records, not contamination of every record.

3. What can the release owner do?

B. Postpone the decision

The briefing explicitly allows postponement; none of the other actions creates the missing evidence.

D Language checks / page 27

1. Which meeting statement is supported?

C. The 94 percent score includes 40 training overlaps.

The overlap is confirmed; no clean-subset score or production outcome has been established.

2. What does "not yet re-scored" mean?

D. The revised calculation remains unfinished.

Not yet states timing and incompleteness, not the direction or final consequence of the result.

3. Which question targets a label dispute?

A. Which rubric criterion distinguishes these two labels?

The rubric provides criteria for adjudication. Predictions, duplicate removal and an aggregate score do not resolve what the reference category should be.

4. Which phrase correctly names the base of a rate?

A. state the denominator

The denominator is the total count against which the number of successes is expressed.

03 Conversation answers

Data, Datasets, Labels, and Leakage

03

E Dialogue gaps

Pages 28-29

1. contamination. Turn 2. Contamination means the evaluation is affected by unwanted exposure or overlap with training data.

2. re-score. Turn 4. Re-scoring calculates performance on the specified remaining records; arithmetic on the overlap rate cannot supply that score.

3. provenance. Turn 6. Provenance records where data came from and how it changed; it can help identify when overlap entered the sets.

4. adjudication. Turn 8. Adjudication resolves disputed labels through the agreed review process rather than selecting whichever label improves the score.

5. denominator. Turn 10. The denominator identifies how many records are included, so exclusions cannot be hidden behind an unchanged headline rate.

6. holdout. Turn 12. A holdout must remain separate from the process being evaluated; the confirmed training overlap compromises that separation.

7. representativeness. Turn 14. Representativeness concerns whether the sample reflects the intended customer population, a separate question from overlap.

8. exact duplicates. Turn 16. Exact duplicates are identical under the chosen comparison rule; verifying them does not complete a near-duplicate review.

9. model selection. Turn 18. Model selection uses evaluation results to choose a model or configuration, so that set no longer provides an untouched final test.

10. limitation statement. Turn 20. A limitation statement specifies what the evidence cannot support, preventing the headline figure from becoming an unjustified release claim.

F Transfer conversation

Page 30

1. overlap

Analyst: "The current test has training overlap, so its independence is compromised."

Ten test messages also appeared in training, which compromises the claimed separation.

2. re-score

Lead: "Report the clean-subset score only after you re-score those 70 messages."

The performance on the remaining 70 cannot be inferred simply by subtracting the overlap count.

3. adjudication

Analyst: "The three disputed labels also need adjudication."

Label disagreements require a defined review decision separate from duplicate removal.

4. denominator

Lead: "And include the denominator with every reported percentage."

A percentage must identify its underlying number of examples, especially after exclusions.

04 Briefing and language checks

Retrieval, Embeddings, Vector Search, and RAG

A Briefing checks / page 31

1. What improved in the fixed test?

A. Queries with a relevant retrieved passage rose from 78 to 86.

The measured gain concerns retrieval relevance on 100 fixed queries; answer quality and user preference remain unchecked.

2. Which component stayed unchanged?

D. The generator

The briefing holds the generator and query set constant, helping isolate the retrieval change.

3. Which claim needs further evidence?

D. The new path improves end-to-end answer quality.

No end-to-end answer review has been completed, so better passage retrieval does not establish that claim.

D Language checks / page 36

1. Which comparison correctly states the observed gain?

B. Coverage increased by eight percentage points on the fixed set.

The measure rises from 78 of 100 to 86 of 100, an eight-percentage-point coverage change.

2. What is a reranker ablation here?

B. Compare the path with and without the reranker while holding other factors fixed.

Changing the reranker alone helps attribute differences in the matched comparison.

3. Which sentence preserves the limitation?

D. Retrieval improved on this set; final-answer quality remains unmeasured.

The sentence separates the measured gain from an unmeasured downstream outcome.

4. Which question checks grounding?

C. Does this passage support the answer's refund exception?

Grounding concerns support for the specific claim. A marker, indexed document or lexical overlap can exist without supporting the stated exception.

04 Conversation answers

Retrieval, Embeddings, Vector Search, and RAG

E Dialogue gaps

Pages 37-38

1. candidate set. Turn 2. The candidate set contains the retrieved items available to the reranker; changes there could affect the comparison.

2. ablation. Turn 4. An ablation compares a component's presence or configuration while controlling other factors, helping isolate its effect.

3. end-to-end. Turn 6. End-to-end review follows the whole request-to-answer path, rather than stopping at the retrieval component.

4. reranker. Turn 8. A reranker can reorder existing candidates; it cannot recover a document absent from the candidate set by reordering alone.

5. chunk boundary. Turn 10. A chunk boundary can separate a main rule from its exception, making a retrieved segment incomplete without neighboring context.

6. grounding. Turn 12. Grounding asks whether the generated claim is supported by the supplied evidence, not merely whether a citation marker appears.

7. index freshness. Turn 14. Index freshness concerns whether indexed material reflects current source information; relevance alone does not establish currency.

8. trade-off. Turn 16. A trade-off involves competing outcomes, here relevance and delay; user acceptance requires evidence beyond the retrieval score.

9. query type. Turn 18. Query type identifies a meaningful subgroup; subgroup timing can reveal costs concealed by an overall average.

10. conditional. Turn 20. Conditional support depends on the stated review being completed; it is not unconditional permission to deploy.

F Transfer conversation

Page 39

1. source

Reviewer: "The citation identifies a real source, but not support for this claim."

Identifying an authentic source does not establish that the cited material supports the particular answer.

2. grounding

Engineer: "Both sections were retrieved, so we should inspect answer grounding."

With both sections available, the question is whether the generated claim reflects the supporting evidence.

3. context

Reviewer: "The exception is missing from the answer, although it is present in the context."

The provided retrieval context contains the exception; the failure is not evidence that it was absent.

4. end-to-end

Engineer: "We will report that distinction in the end-to-end review."

The full review follows the evidence through retrieval into the final answer, preserving where the mismatch appears.

04

05 Briefing and language checks

Fine-Tuning, Alignment, and Adaptation

A Briefing checks / page 40

05

1. Which fact directly concerns outdated information?

A. The retrieval index contains obsolete policy wording.

The index supplies obsolete source wording, which is a distinct issue from output formatting.

2. What is known about the 200 examples?

D. They are permission-cleared but consistency is unaudited.

Permission to use data does not establish its consistency, currency, or training value.

3. What distinction should the experiment preserve?

D. Source freshness versus output-format behavior

The two observed failures concern source currency and output behavior. The alternatives conflate separate checks or assume an untested mechanism.

D Language checks / page 45

1. Which action directly addresses the known stale source?

D. Correct the retrieval index through the approved update process.

The obsolete wording is in retrieval; a source update addresses that known mechanism without assuming training will refresh it.

2. Which is a behavioral target?

D. Returning the required fields consistently

Format adherence is an observable output behavior; the other items do not specify model behavior.

3. Which claim needs qualification?

A. Fine-tuning will solve both failures.

The proposed remedy has not been evaluated and does not directly maintain source freshness.

4. Which comparison best isolates the formatting intervention?

C. Keep sources and test cases fixed while changing the formatting method.

Holding sources and cases fixed makes the effect of the formatting change easier to interpret.

05 Conversation answers

Fine-Tuning, Alignment, and Adaptation

E Dialogue gaps

Pages 46-47

- 1. failure modes.** Turn 2. Failure modes identify distinct ways a system fails; wrong structure and stale factual content need not share a cause.
- 2. behavioral target.** Turn 4. A behavioral target specifies the output behavior to change, here adherence to the required fields.
- 3. consistency.** Turn 6. Consistency means examples follow compatible criteria; permission alone says nothing about agreement among labels or formats.
- 4. few-shot.** Turn 8. Few-shot prompting supplies a small set of examples at inference time, providing a useful comparison with parameter updates.
- 5. format adherence.** Turn 10. Format adherence measures conformity to the required structure, which must remain distinct from factual correctness.
- 6. schema.** Turn 12. A schema defines the required structure and allowed fields; it does not authorize inventing unsupported values.
- 7. generalization.** Turn 14. Generalization concerns performance beyond the examples used to fit or choose the method.
- 8. catastrophic forgetting.** Turn 16. Catastrophic forgetting names loss of previously learned capabilities during later training; a targeted gain does not rule it out.
- 9. rollback candidate.** Turn 18. A rollback candidate is a previously checked configuration available if the new adaptation must be withdrawn.
- 10. hypotheses.** Turn 20. Hypotheses are proposed explanations or outcomes to test; they are not established findings before the comparison.

F Transfer conversation

Page 48

1. category list

Scientist: "The prompt supplies an obsolete category list, so we should correct that input first."

The known discrepancy is the six-category instruction, not evidence that a training change is already required.

2. baseline

Lead: "Then compare the corrected prompt with the existing baseline."

A baseline provides the reference against which the effect of the corrected prompt can be measured.

3. held-out

Scientist: "We will measure adherence on held-out examples, not only the demonstrations."

Examples kept outside fitting and selection provide a more independent check than examples used to demonstrate the task.

4. hypothesis

Lead: "Until then, the benefit of further training remains a hypothesis."

No training comparison has been completed, so its benefit is a proposed outcome rather than a finding.

05

06 Briefing and language checks

Evaluation, Benchmarks, and Regression

A Briefing checks / page 49

1. What happens to total passes?

C. They fall from 90 to 88.

The 12-to-18 improvement applies only to long documents; total passes decline by two.

2. Which slice improves?

A. Long-document cases

Long-document passes rise from 12 to 18; short-request passes fall from 78 to 70.

3. What remains unknown?

C. The production mixture and seriousness of failures

The briefing supplies test counts and an unchanged rubric, but not production proportions or severity analysis.

D Language checks / page 54

1. Which summary preserves both findings?

D. Total passes fell by two; long-document passes rose by six.

The statement correctly separates the overall count from the improved subgroup and makes no production claim.

2. What is the candidate's long-document pass rate?

A. 90 percent

Eighteen divided by twenty equals 90 percent; 88 percent is the aggregate test rate.

3. Which question targets practical significance?

C. How serious are the newly failing cases for users?

Consequences for users address practical importance. Stability, arithmetic reconciliation and rubric consistency are also useful, but answer different questions.

4. Which proposed conclusion exceeds the evidence?

C. Long-document routing is already safe to release.

No routing or severity assessment has established that a specialized release is acceptable.

06 Conversation answers

Evaluation, Benchmarks, and Regression

E Dialogue gaps

Pages 55-56

- 1. aggregate.** Turn 2. The aggregate combines all test cases, so it prevents a favorable subgroup from standing in for the whole result.
- 2. regression.** Turn 4. A regression is a worsening of measured behavior; here eight fewer short-request cases pass.
- 3. acceptance criteria.** Turn 6. Acceptance criteria are the conditions for approving a version, ideally set before selecting whichever metric looks favorable.
- 4. paired comparison.** Turn 8. A paired comparison examines corresponding cases across versions, revealing which individual outcomes changed rather than only comparing totals.
- 5. severity.** Turn 10. Severity describes how consequential a failure is; equal pass-fail counts need not imply equal user impact.
- 6. human review.** Turn 12. Human review applies the stated criteria to actual outputs and can examine judge disagreements in the changed subgroup.
- 7. false positives.** Turn 14. False positives in the proposed router would send requests into the long-document path when they do not meet its intended condition.
- 8. statistical significance.** Turn 16. Statistical significance concerns evidence under a statistical model; it does not alone establish that a difference matters operationally.
- 9. disaggregation.** Turn 18. Disaggregation separates the combined result into defined groups, making different patterns visible without discarding the overall result.
- 10. release gate.** Turn 20. A release gate is the decision checkpoint whose criteria must be met; presenting an improved slice does not automatically satisfy it.

F Transfer conversation

Page 57

1. aggregate

Reviewer: "The overall aggregate hides a weak urgent-message subgroup."

A combined score can conceal poor results in a smaller but important subgroup.

2. false negatives

Engineer: "Those missed urgent messages are false negatives for that category."

The messages are urgent but the classifier fails to identify them as positive urgent cases.

3. severity

Reviewer: "We need a separate assessment of their severity."

The seriousness of missing urgent messages cannot be inferred from the overall success count.

4. release gate

Engineer: "Then the release gate should remain unresolved until that review is complete."

The approval checkpoint lacks relevant evidence and should not be treated as passed.

06

07 Briefing and language checks

Inference, Latency, Cost, and Deployment

A Briefing checks / page 58

1. Which result is measured?

C. Mean response time falls from four seconds to two.

The case reports a mean on a fixed test, not a guarantee for every request or an unmeasured workload.

2. What traffic is permitted in the internal pilot?

A. Synthetic requests only

The stated pilot boundary allows synthetic requests only; broader traffic needs separate authorization.

3. Which responsibility is unresolved?

D. Who owns rollback

The case names operational and product owners and a previous configuration, but no rollback owner.

D Language checks / page 63

1. Which claim is supported by the test?

B. Mean latency decreased by 50 percent.

The mean falls from four to two, a reduction of half; the other claims concern unmeasured outcomes.

2. Which question distinguishes first visible output from a finished response?

C. Is that first-token time or complete-answer time?

First-token and completion times identify the two user-visible events in the question. The other comparisons describe distributions or different serving measures.

3. Which statement preserves authorization boundaries?

D. The synthetic pilot does not authorize production traffic.

The case explicitly separates the internal pilot from production approval.

4. Which expression means restoring the earlier configuration?

B. initiate rollback

Rollback returns the system to a previous configuration; the other actions change different aspects of serving.

07 Conversation answers

Inference, Latency, Cost, and Deployment

E Dialogue gaps

Pages 64-65

- 1. tail latency.** Turn 2. Tail latency describes the slower end of the response-time distribution, which an improved average does not establish.
- 2. mean.** Turn 4. The mean is an arithmetic average; it is not an upper bound on individual request times.
- 3. time to first token.** Turn 6. Time to first token measures the initial wait, not the time needed to receive and verify the whole answer.
- 4. concurrency.** Turn 8. Concurrency is the amount of simultaneous work; varying it helps expose waiting and resource contention.
- 5. fallback.** Turn 10. A fallback is an alternate serving path; its extra work can affect both cost and observed response time.
- 6. rollback.** Turn 12. Rollback restores a previous configuration; availability of that configuration does not establish who is authorized to restore it.
- 7. stop condition.** Turn 14. A stop condition states when the pilot should pause, avoiding an improvised decision after a concerning result appears.
- 8. rollout.** Turn 16. A rollout introduces the configuration to its intended users; it requires its own scope and approval, beyond this synthetic test.
- 9. queueing.** Turn 18. Queueing is waiting before processing resources are available; it can dominate latency even when model inference itself is faster.
- 10. limitations.** Turn 20. Limitations identify the conditions and missing evidence that constrain how widely a measured result can be applied.

F Transfer conversation

Page 66

1. time to first token

Engineer: "One second is the time to first token, not the time for the full response."

The first visible token arrives at one second; the full answer takes six seconds.

2. completion time

Lead: "Then show completion time separately so the comparison is clear."

Completion time captures the wait for the entire answer and must not be replaced by the first-token measure.

3. service target

Engineer: "Neither figure is an approved service target."

The scenario explicitly says no target has been approved; measurements do not themselves authorize a promise.

4. qualification

Lead: "Keep that qualification attached when the slide is shared."

The limiting explanation prevents readers from treating a one-second initial display as full-answer performance.

08 Briefing and language checks

Safety, Security, Privacy, and Governance

A Briefing checks / page 67

1. What has been confirmed?

B. One reproduction in an authorized synthetic test

The case confirms one synthetic-test reproduction and explicitly leaves broader scope unknown.

2. Where should restricted evidence go?

B. The approved restricted reporting channel

The case provides an approved restricted route; the other destinations expand access unnecessarily.

3. Which distinction must the report preserve about impact?

C. Synthetic exposure versus real-customer impact

The test uses fictional data and one tested configuration. It demonstrates the supplied behavior without establishing real-customer exposure or wider scope.

D Language checks / page 72

1. Which headline matches the known scope?

B. Synthetic secret repeated in one authorized test; broader scope unconfirmed.

The supported headline retains both the observed reproduction and its limited scope. Synthetic data do not erase the behavior, and a proposed control is not a verified resolution.

2. Which statement distinguishes evidence from intent?

D. The document attempted redirection, and the output repeated the synthetic secret.

The supported statement describes observable content and output without inventing motives for people or the model.

3. What is a mitigation?

B. A measure intended to reduce risk or impact

A mitigation reduces risk; effectiveness and remaining risk still require evaluation.

4. Which wording avoids an unsupported absence claim?

A. We have not established real-customer impact.

The correct statement reports the current evidence boundary without claiming that all possible impact has been excluded.

08 Conversation answers

Safety, Security, Privacy, and Governance

E Dialogue gaps

Pages 73-74

1. restricted channel. Turn 2. The restricted channel limits access to detailed evidence under the team's approved reporting process.

2. prompt-injection. Turn 4. Prompt injection refers to untrusted material attempting to redirect model behavior; the report must still distinguish the attempt from the observed result.

3. scope limitation. Turn 6. A scope limitation states which environments or cases have and have not been assessed, preventing unsupported generalization.

4. containment. Turn 8. Containment concerns limiting further consequences while investigation proceeds; its owner must make the applicable operational decision.

5. audit trail. Turn 10. An audit trail preserves the records needed to reconstruct actions, configuration, and review decisions.

6. mitigation. Turn 12. A mitigation is a control intended to reduce the risk; one successful replay test does not establish that every relevant variant is controlled.

7. permission scope. Turn 14. Permission scope identifies which resources or actions were authorized, helping reviewers assess the boundaries relevant to the finding.

8. redaction. Turn 16. Redaction removes restricted detail from a shared copy while preserving necessary original evidence through approved handling.

9. residual risk. Turn 18. Residual risk is what remains after a control is applied; pending checks prevent a blanket claim that the risk is gone.

10. finding. Turn 20. A finding records the observed issue and its review status; it should remain open until the authorized process accepts sufficient evidence.

F Transfer conversation

Page 75

1. redacted

Tester: "I will share a redacted summary in the project chat."

The broad audience needs the finding's meaning without unnecessary credential or configuration details.

2. restricted

Reviewer: "Place the original screenshot in the approved restricted repository."

Restricted access preserves the detailed evidence for authorized review rather than widening distribution.

3. scope

Tester: "The update will state the tested environment and the unknown scope."

The report must separate the assessed environment from configurations whose behavior has not been established.

4. safety

Reviewer: "And do not call a proposed detector a guarantee of safety."

A proposed control may reduce risk, but its effectiveness and remaining limitations require evaluation.

08

09 Conversation answers

The annotators disagree about what counts as a request

E Extended conversation

Pages 76-77

1. rubric. Turn 5. Rubric means the written criteria for assigning labels, which are unclear for this message.

2. boundary case. Turn 8. A boundary case sits near the dividing line between categories and helps clarify the rule.

3. adjudication. Turn 12. Adjudication is a defined review process for resolving disputed labels, not simply accepting the first opinion.

4. calibration sample. Turn 16. A calibration sample lets annotators practice and compare judgments under the revised instructions.

5. dataset version. Turn 19. Dataset version identifies the particular release of examples and labels used in an experiment.

6. release status. Turn 20. Release status states whether the data are approved for use, rather than merely present in storage.

F Transfer exchange

Page 78

1. guideline

Annotator: This message doesn't fit either example in the current guideline.

Guideline is the written labeling instruction the annotator has consulted.

2. review

Lead: Add it to the review queue with the reason for uncertainty.

Review identifies the route for a judgment that needs further examination.

3. label

Annotator: Should I preserve the original text and my provisional label?

Label is the tentative category assignment, distinct from the source message.

4. criteria

Lead: Yes. We'll record the final decision and any change to the criteria.

Criteria are the rules used to assign categories consistently across examples.

10 Conversation answers

The training job stopped after the last checkpoint

E Extended conversation

Pages 79-80

- 1. checkpoint.** Turn 2. Checkpoint means saved training state from a particular point, not necessarily the latest exported weights.
- 2. optimizer state.** Turn 6. Optimizer state stores training-update information beyond model weights, such as accumulated momentum values.
- 3. learning-rate schedule.** Turn 10. Learning-rate schedule controls how the update rate changes as training progresses.
- 4. reproducibility.** Turn 14. Reproducibility concerns obtaining consistent results under documented conditions, not a universal guarantee of identical weights.
- 5. queue wait.** Turn 18. Queue wait is time awaiting available resources, not time spent actively training the model.
- 6. resume point.** Turn 20. Resume point identifies the saved state from which the intended training process can continue.

F Transfer exchange

Page 81

1. weights

Researcher: Both runs began with the same model weights, so I called them identical starts.

Weights are shared, but optimizer and scheduler state can still differ.

2. condition

Lead: One run used a fresh optimizer. Record that as a changed condition.

Condition identifies an experimental difference that affects interpretation of the comparison.

3. reinitialized

Researcher: I'll distinguish the resumed run from the reinitialized run.

Reinitialized describes the run that reset training state instead of fully resuming it.

4. comparison

Lead: Good. Keep both results, but qualify the comparison.

Comparison must acknowledge the different starting states rather than attributing all differences elsewhere.

11 Conversation answers

A drafting assistant is about to become a sending agent

E Extended conversation

Pages 82-83

1. permission scope. Turn 2. Permission scope defines which actions and resources the assistant is authorized to use.

2. human approval. Turn 6. Human approval is an explicit decision by a person about the proposed action, not merely launching the task.

3. Least privilege. Turn 12. Least privilege limits available permissions to those needed for the defined task.

4. sandbox. Turn 14. Sandbox means an isolated test environment designed to prevent effects on real customer accounts.

5. audit log. Turn 16. Audit log is a record of actions and approvals used to reconstruct what happened.

6. kill switch. Turn 20. Kill switch is a control that can stop the agent workflow promptly when needed.

F Transfer exchange

Page 84

1. content

Operator: The content no longer matches what I reviewed.

Content is the message text, which changed after the operator's approval.

2. send

Engineer: Then the previous approval must not authorize this send.

Send is the consequential action that must match the reviewed message.

3. draft

Operator: I'll request a fresh review of the exact recipient and draft.

Draft is the proposed message that the reviewer must see before sending.

4. outcome

Engineer: Record the replacement approval and the action's final outcome.

Outcome records whether the approved action succeeded, failed, or remained unresolved.

11

Find the move you need

Use these as precise language models. Replace details only when you know the actual facts, authority, and agreed conditions.

01 Speaking the AI Development Stack

Locate the layer: Which component first shows the wrong value?

Request a trace: Can you attach the request ID and the redacted trace?

Agree on a handoff: I will send the evidence and name the unresolved question.

Full phrase set: pages 7-8 | Dialogue: pages 10-11

02 LLMs, Transformers, Tokens, and Context

Restate a proposal: You are proposing a larger output allowance, correct?

Correct tactfully: I see the aim, but that change would increase the reservation.

State a condition: A smaller output allowance is acceptable only if the summary still meets the task.

Full phrase set: pages 16-17 | Dialogue: pages 19-20

03 Data, Datasets, Labels, and Leakage

Flag compromised evidence: The headline score includes records that overlap with training.

Reject unsupported subtraction: Removing eight percent of records does not tell us the new accuracy.

Offer a bounded update: I can provide the overlap list before the meeting.

Full phrase set: pages 25-26 | Dialogue: pages 28-29

04 Retrieval, Embeddings, Vector Search, and RAG

Name the measure: The improvement is passage coverage on this fixed query set.

Request a matched comparison: Can we hold the generator and queries constant?

Define the next test: Review final answers and timing on the same cases.

Full phrase set: pages 34-35 | Dialogue: pages 37-38

Keep the conversation productive

Use these as precise language models. Replace details only when you know the actual facts, authority, and agreed conditions.

05 Fine-Tuning, Alignment, and Adaptation

Separate failures: We have a formatting issue and a source-freshness issue.

Propose a baseline: Let us test clearer prompting before attributing gains to training.

Agree on evidence: We will compare format adherence and factual correctness separately.

Full phrase set: pages 43-44 | Dialogue: pages 46-47

06 Evaluation, Benchmarks, and Regression

Lead with both results: Overall passes fell, while the long-document slice improved.

Request paired cases: Which previously passing cases now fail?

Propose the next review: Compare the changed cases and classify their failure severity.

Full phrase set: pages 52-53 | Dialogue: pages 55-56

07 Inference, Latency, Cost, and Deployment

State the measured gain: Mean response time fell from four seconds to two on this test.

Separate timings: Is that time to first token or time to complete the answer?

Preserve a distinction: Pilot approval does not authorize production rollout.

Full phrase set: pages 61-62 | Dialogue: pages 64-65

08 Safety, Security, Privacy, and Governance

Lead with the observation: In one authorized test, the assistant repeated the synthetic secret.

Separate intent and result: The document attempted to redirect behavior; this output is what we observed.

Avoid premature closure: A blocked replay is encouraging, but it does not prove the issue is resolved.

Full phrase set: pages 70-71 | Dialogue: pages 73-74

Find a term and its context

The page number leads to the definition and a typical word combination. Repeated terms have separate entries when they serve different lesson contexts.

ablation run a reranker ablation	33
acceptance criterion agree on acceptance criteria	50
adapter load a task adapter	41
adjudication request label adjudication	24
aggregate report the aggregate score	50
alignment define the alignment objective	41
annotation review annotation quality	23
API document an API contract	6
application stack map the application stack	5
attack surface map the attack surface	69
attention compute attention weights	14
audit trail preserve the audit trail	69
autoscaling review autoscaling behavior	60
base model compare against the base model	41
baseline establish a baseline	50
batching evaluate dynamic batching	59
behavioral target define the behavioral target	42
benchmark interpret a benchmark result	50
calibration assess confidence calibration	51
canary define a canary population	60
candidate evaluate the candidate	50
candidate set inspect the candidate set	33
capacity planning review capacity assumptions	60
catastrophic forgetting assess catastrophic forgetting	42

Find a term and its context

The page number leads to the definition and a typical word combination. Repeated terms have separate entries when they serve different lesson contexts.

checkpoint select a checkpoint	15
chunk retrieve a relevant chunk	32
chunk boundary preserve context at chunk boundaries	32
citation verify a citation	33
class imbalance check class imbalance	24
cold start measure cold-start effects	60
compression assess compression losses	15
concurrency test concurrent requests	59
confidence interval report a confidence interval	51
confusion matrix examine the confusion matrix	51
containment confirm containment ownership	69
contamination quantify evaluation contamination	23
context window stay within the context window	14
contract preserve the service contract	6
cost per request compare cost per request	60
credential avoid exposing credentials	68
data exfiltration assess exfiltration risk	68
data leakage investigate data leakage	23
dataset curate a dataset	23
deduplicate deduplicate before splitting	24
deduplication apply exact deduplication	15
delimiter preserve clear delimiters	15
denominator state the denominator	24
dependency identify a service dependency	5

Find a term and its context

The page number leads to the definition and a typical word combination. Repeated terms have separate entries when they serve different lesson contexts.

disaggregation	disaggregate by task type	50
disclosure	coordinate responsible disclosure	69
distribution shift	detect distribution shift	24
downstream	assess downstream impact	5
DPO	evaluate a DPO experiment	41
embedding	compute an embedding	32
end-to-end	measure end-to-end latency	33
endpoint	call an endpoint	5
epoch	record the epoch count	42
error budget	track error-budget consumption	60
evaluation	design an evaluation	50
exact duplicate	remove exact duplicates	23
exception clause	retain the exception clause	15
fallback	test the fallback path	60
false negative	review false negatives	51
false positive	inspect false positives	51
few-shot prompting	test few-shot prompting	42
filter	apply a metadata filter	33
fine-tuning	evaluate a fine-tuning run	41
format adherence	measure format adherence	42
generalization	test generalization	42
golden set	maintain a golden set	50
governance	clarify governance responsibilities	69
grounding	check answer grounding	33

Find a term and its context

The page number leads to the definition and a typical word combination. Repeated terms have separate entries when they serve different lesson contexts.

guardrail evaluate a guardrail	68
handoff prepare a clear handoff	6
headroom leave sufficient headroom	15
holdout maintain a clean holdout	23
human oversight define human-oversight responsibilities	69
human review conduct a blinded human review	51
hybrid retrieval test hybrid retrieval	32
hyperparameter document a hyperparameter choice	42
incident report file an incident report	69
index freshness monitor index freshness	33
inference run inference	5
input tokens measure input tokens	14
instruction hierarchy respect the instruction hierarchy	15
integration test the integration	6
inter-rater agreement measure inter-rater agreement	24
keyword search combine keyword search with vectors	32
label verify a disputed label	23
language model query a language model	5
latency measure response latency	59
learning rate compare learning-rate settings	42
least privilege apply least-privilege access	68
LLM evaluate an LLM	14
LLM-as-judge calibrate an LLM-as-judge setup	51
load test run a controlled load test	60

Find a term and its context

The page number leads to the definition and a typical word combination. Repeated terms have separate entries when they serve different lesson contexts.

long-context	evaluate long-context performance	15
LoRA	train a LoRA adapter	41
mean	report mean latency	59
median	compare median response times	59
mitigation	evaluate a proposed mitigation	69
near duplicate	review near duplicates	23
observability	improve observability	6
orchestration	trace the orchestration flow	5
output tokens	reserve output tokens	14
overfitting	monitor for overfitting	42
overlap window	adjust the overlap window	32
P95	inspect P95 latency	59
paired comparison	run a paired comparison	51
parameter	update model parameters	14
pass rate	report the pass rate	50
payload	inspect the request payload	5
permission scope	document permission scope	68
PII	minimize PII in evidence	68
position	vary evidence position	15
practical significance	assess practical significance	51
precision	improve retrieval precision	33
preference data	audit preference data	41
pretraining	distinguish pretraining from adaptation	41
prompt	revise the prompt	14

Find a term and its context

The page number leads to the definition and a typical word combination. Repeated terms have separate entries when they serve different lesson contexts.

prompt injection	report a prompt-injection attempt	68
provenance	establish data provenance	24
queueing	measure queueing delay	59
RAG	evaluate a RAG pipeline	32
rate limit	respect a rate limit	60
re-score	re-score the clean subset	24
recall	measure retrieval recall	33
red teaming	define a red-team scope	68
redaction	prepare a redacted report	68
regression	investigate a regression	50
release gate	satisfy the release gate	51
relevance	judge passage relevance	33
representativeness	assess sample representativeness	24
reproduction	obtain a reliable reproduction	6
reproduction steps	submit restricted reproduction steps	69
request ID	attach the request ID	6
reranker	evaluate the reranker	32
reserved capacity	reduce reserved capacity	15
residual risk	document residual risk	69
retention	follow the evidence-retention policy	69
retrieval	inspect retrieval results	5
RLHF	discuss an RLHF workflow	41
rollback	assign rollback ownership	60
rollback candidate	retain a rollback candidate	42

Find a term and its context

The page number leads to the definition and a typical word combination. Repeated terms have separate entries when they serve different lesson contexts.

rollout stage the rollout	60
root cause verify the root cause	6
rubric apply a shared rubric	24
sampling frame define the sampling frame	24
schema validate a response schema	6
secret protect a test secret	68
semantic search assess semantic search	32
severity assess failure severity	51
SFT compare SFT with prompting	41
similarity search run similarity search	32
slice analyze a performance slice	50
SLO agree on an SLO	60
source attribution preserve source attribution	33
split define a reproducible split	23
statistical significance distinguish significance from usefulness	51
stratification use stratified sampling	24
streaming enable response streaming	59
structured output validate structured output	42
supervised fine-tuning prepare supervised fine-tuning data	41
symptom describe the symptom	6
system instruction inspect the system instruction	14
tail latency investigate tail latency	59
test case inspect a failed test case	50
test set preserve the test set	23

Find a term and its context

The page number leads to the definition and a typical word combination. Repeated terms have separate entries when they serve different lesson contexts.

threat model	review the threat model	69
throughput	measure requests per second	59
time to first token	track time to first token	59
token	count input tokens	14
token budget	allocate a token budget	14
token rate	measure output token rate	59
tokenizer	use the model's tokenizer	14
tool call	inspect a tool call	6
top-k	compare top-k results	33
trace	examine a request trace	5
training artifact	version the training artifact	42
training objective	specify the training objective	41
training set	audit the training set	23
transformer	use a transformer architecture	14
triage	triage the defect	6
truncation	detect silent truncation	15
trust boundary	identify a trust boundary	68
untrusted content	isolate untrusted content	68
upstream	inspect the upstream response	5
user interface	inspect the user interface	5
validation set	monitor validation performance	23
vector	compare document vectors	32
vector store	query a vector store	32
weight	load model weights	15

Original cases. Clear boundaries.

These are original fictional teaching conversations, not transcripts of real workers. Names, figures, organizations, and incidents are invented. Sources support terminology and communication aims, not the fictional results or an endorsement of this book.

[1] NIST. Artificial Intelligence Risk Management Framework: Generative AI Profile (2024).

[Open the original source](#)

Background for the vocabulary of evaluation, information integrity, privacy, and documented risk decisions. The fictional release criteria are not NIST requirements. Checked 10 October 2026.

[2] Lewis et al. Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks (2020).

[Open the original source](#)

Primary research background for retrieval-augmented generation. The book does not reproduce the paper or present its results as product guarantees. Checked 10 October 2026.

[3] PyTorch. Saving and Loading Models.

[Open the original source](#)

Background for distinguishing model weights from the additional state needed to resume training. Exact recovery requirements depend on the training system. Checked 10 October 2026.

[4] OWASP. LLM06:2025 Excessive Agency.

[Open the original source](#)

Background for bounded tool permissions, independent authorization, and human review of consequential agent actions. Checked 10 October 2026.

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